Industrial Internship Report

Tech Elecon Pvt. Ltd

Submitted by

Om Patel

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In partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

Computer Science and Design

G H Patel College of Engineering & Technology

The Charutar Vidya Mandal (CVM) University,

Vallabh Vidyanagar - 388120

May 2026

GCET
G H PATEL COLLEGE OF ENGINEERING & TECHNOLOGY

CVM UNIVERSITY

G H Patel College of Engineering & Technology

Bachelor of Technology (Computer Science and Design)

CERTIFICATE

This is to certify that **Om Patel (12202130501047)** has submitted the Industrial Internship report based on internship undergone at **Tech Elecon Pvt. Ltd.** for a period of **16** weeks from **1st January 2026** to **30th April 2026** in partial fulfillment for the degree of Bachelor of Technology in **Computer Science and Design, G H Patel College of Engineering and Technology** at The Charutar Vidya Mandal (CVM) University, Vallabh Vidyanagar during the academic year 2025-26.

G. H. PATEL COLLEGE OF ENGINEERING & TECHNOLOGY
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Dr. Kinjal Joshi
Internal Guide

Dr. Sudhir Vegad
Head of the Department

TECHELECON
TECHNOLOGY SIMPLIFIED

HR/TEPL/634 30.04.2026

CERTIFICATE

*This is to certify that **Mr.Om Patel** Student of **Computer Engineering** from **G H Patel College of Engineering & Technology – V V Nagar** has undergone for **Internship** in our organization and completed successfully during **01.01.2026** to **30.04.2026.***

During this period, we have found him regular, sincere and enthusiastic.

For, Tech Elecon Pvt. Ltd.

Nirali Trivedi
Group HR Head

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DECLARATION

I, Om Patel (12202130501047), hereby declare that the Industrial Internship report submitted in partial fulfillment for the degree of Bachelor of Technology in Computer Science and Design, G H Patel College of Engineering & Technology, The Charutar Vidya Mandal (CVM) University, Vallabh Vidyanagar, is a bonafide record of work carried out by me at Tech Elecon Pvt. Ltd. under the supervision of Mr. Satyam Raval and Dr. Kinjal Joshi and that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

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Om Patel

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The completion of this work would not have been possible without the cooperation, coordination, and support of several individuals. I would like to express my sincere gratitude to **Mr. Satyam Raval**, my industry mentor, for his valuable guidance, encouragement, and continuous support throughout the duration of my internship.

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I am also grateful to **all the faculty members** of the Department of Computer Engineering, G H Patel College of Engineering and Technology, Vallabh Vidyanagar**,** for their guidance and support.

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ABSTRACT

This report presents GearMind AI, a full-stack industrial gear fault detection and predictive maintenance dashboard developed during a 16-week internship at Tech Elecon Pvt. Ltd., Anand, Gujarat.

GearMind AI employs a per-gear-type multi-model architecture with dedicated classifiers trained for each of the four gear types: Gradient Boosting Machine (GBM) for helical gear (99.87% accuracy), SVM with RBF kernel for spur gear (82.3%), SVM with RBF kernel for bevel gear (99.4%), and Gradient Boosting for worm gear (100% accuracy on the 50,000-sample synthetic dataset). Each gear type also has a dedicated RUL regressor, StandardScaler, and label encoder, trained on physics-informed synthetic sensor data with SMOTE oversampling to handle class imbalance. The overall system integrates SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) explainability frameworks to ensure transparent, interpretable predictions suitable for maintenance engineers without deep ML expertise.

The system is built on a React 19 frontend with an eight-module interactive dashboard and a FastAPI backend exposing sixteen REST API endpoints backed by a SQLite database. An AI Copilot powered by LLaMA 3.3 70B via the Groq API provides natural language decision support to maintenance personnel. Additional modules include a Manufacturing QC checker with AGMA-grade compliance, Vibration and PHM analysis with FFT spectral decomposition, a Differential Evolution-based parameter optimizer, and an AI-generated seven-section maintenance report with PDF export capability.

The manufacturing QC module achieved a 96% parameter pass rate during validation. The system maintains a sub-300ms API response time for full prediction plus SHAP computation. Cost-benefit analysis demonstrates savings of Rs. 4.05 to 4.68 Lakh per gear unit when early fault detection replaces failure-driven maintenance. This project demonstrates the practical applicability of explainable AI, full-stack ML system development, and industrial IoT concepts in a real-world manufacturing context.

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Abbreviations

AI	Artificial Intelligence
ML	Machine Learning
XAI	Explainable Artificial Intelligence
GBM	Gradient Boosting Machine
XGB	XGBoost — Extreme Gradient Boosting
RF	Random Forest
SVM	Support Vector Machine
LR	Logistic Regression
SHAP	SHapley Additive exPlanations
LIME	Local Interpretable Model-agnostic Explanations
LLM	Large Language Model
RUL	Remaining Useful Life
PHM	Prognostics and Health Management
FFT	Fast Fourier Transform
RMS	Root Mean Square
MTBF	Mean Time Between Failures
AGMA	American Gear Manufacturers Association
API	Application Programming Interface
REST	Representational State Transfer
RBAC	Role-Based Access Control
QC	Quality Control
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
SDLC	Software Development Life Cycle
SQLite	Self-Contained Serverless SQL Database Engine
IoT	Internet of Things
SMOTE	Synthetic Minority Over-sampling Technique
JWT	JSON Web Token
CRUD	Create, Read, Update, Delete
MLflow	Machine Learning Lifecycle Platform
GBM	Gradient Boosting Machine

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1. OVERVIEW OF THE COMPANY

1.1 HISTORY OF TECH ELECON PVT. LTD.

Tech Elecon Private Limited is a private limited company based in Vallabh Vidyanagar, Gujarat, India. It was incorporated on **30 May 2012** and operates in the field of **information technology and communication services**. The company is involved in providing **IT consulting, software development, networking, and communication solutions**. Over the years, it has developed expertise in delivering integrated solutions combining hardware, software, and networking technologies. Tech Elecon serves a wide range of clients including **industrial, government, and enterprise sectors**, offering solutions that support communication systems and digital infrastructure.

1.2 PRODUCT PORTFOLIO AND SCOPE OF WORK

Tech Elecon Pvt. Ltd. provides a combination of IT services, telecom solutions, and software development services. Its major offerings include:

- **Software Development Services:** Custom software, web applications, and enterprise solutions using technologies such as .NET, PHP, and open-source platforms.
- **Networking & Communication Solutions:** High-speed internet, IP VPN, bandwidth services, and secure data communication systems.
- **Telecom & Infrastructure Services:** VoIP systems, leased line installation, and cloud-based telephony solutions for businesses.
- **Data & Security Solutions:** Data storage, system integration, and secure communication networks.

Table 1.1 Tech Elecon Product Portfolio

Division	Product Category	Applications
Software Division	Custom Software & Web Apps	Business automation, enterprise systems
Networking	Internet & VPN Services	Secure communication, connectivity
Telecom Solutions	VoIP & Cloud Telephony	Business communication systems
IT Services	Data Storage & Security	Data management and protection

1.3 ORGANIZATION CHART

Tech Elecon Pvt. Ltd. follows a structured organizational hierarchy consisting of top management, followed by departmental heads managing various functions such as IT services, networking, software development, and operations. The company includes teams working on:

- Software development
- Network infrastructure
- Telecom services
- Client support and maintenance

The **IT Department**, where this internship was carried out, focuses on software development, system integration, and technical support.

Fig 1.1 Tech Elecon Organization Chart

1.4 CAPACITY OF PLANT

As an IT and telecom solutions provider, Tech Elecon Pvt. Ltd. operates on a **service-based capacity model** rather than manufacturing capacity. The company is capable of handling multiple projects simultaneously in areas such as:

- Software development
- Network setup and maintenance
- Telecom infrastructure deployment

It utilizes modern tools, communication technologies, and technical expertise to deliver scalable and efficient solutions to clients across different sectors.

2. OVERVIEWS OF DEPARTMENTS AND PRODUCTION LAYOUT

2.1 DEPARTMENTS AT TECH ELECON PVT. LTD.

Tech Elecon Pvt. Ltd. is organized into various departments, each contributing to its service delivery:

- **IT & Software Development Department:** Responsible for application development, coding, and system design.
- **Networking & Telecom Department:** Handles internet services, VPN setup, VoIP systems, and communication infrastructure.
- **Data & Security Department:** Manages data storage, cybersecurity, and secure communication systems.
- **Support & Maintenance Department:** Provides technical support, troubleshooting, and system updates.

Operations & Management: Oversees project execution, client coordination, and business operations.

2.2 TECHNICAL SPECIFICATIONS OF MAJOR EQUIPMENT

Table 2.1 Major Equipment at Tech Elecon – Technical Specifications

System / Tool	Specification	Purpose
Software Technologies	.NET, PHP, Open Source	Application development
Networking Systems	IP VPN, Broadband	Connectivity solutions
Telecom Tools	VoIP, Cloud Telephony	Communication systems
Data Systems	Data Storage & Security	Data management

IT Infrastructure	Servers & Network Devices	System deployment

2.3 SCHEMATIC LAYOUT OF PRODUCTION PROCESS

Tech Elecon Pvt. Ltd. follows a **service-based operational workflow**. The process begins with **client requirement analysis**, followed by **system design and planning**. Based on requirements, the company provides solutions such as software development, networking setup, or telecom infrastructure implementation. After development and configuration, the system undergoes **testing and validation** to ensure performance and reliability. The final solution is then **deployed at the client site**, followed by **continuous monitoring, support, and maintenance**. This workflow ensures efficient delivery of services and long-term client satisfaction.

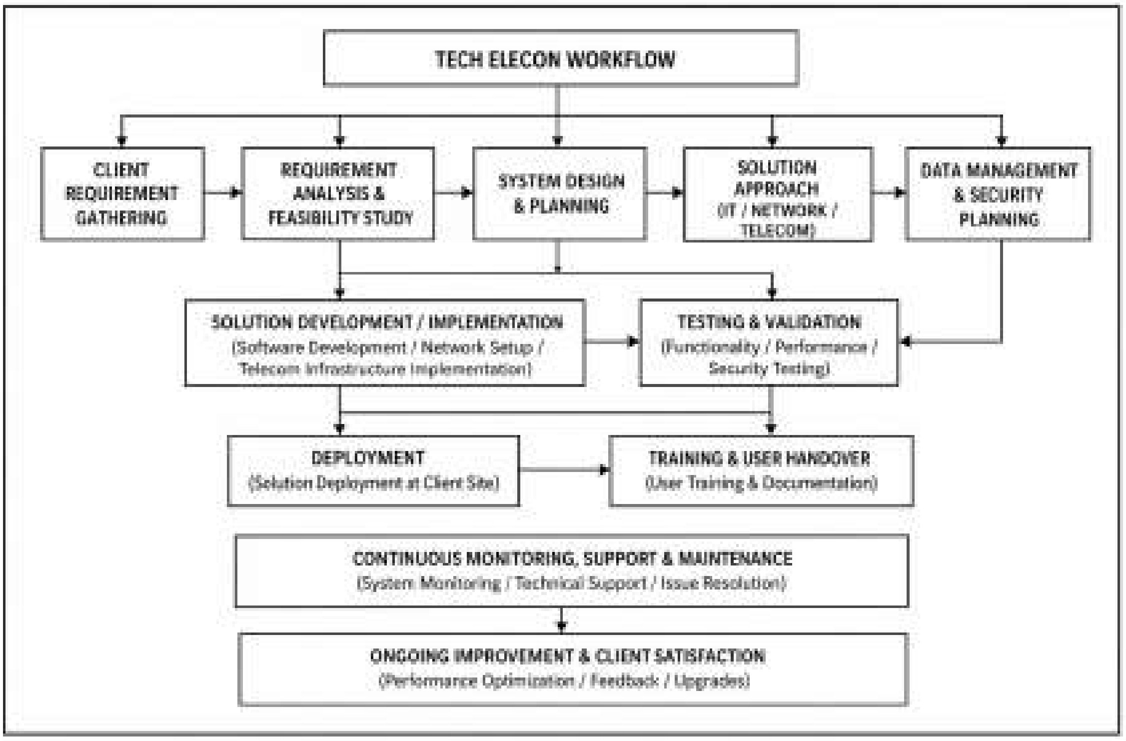


Fig 2.1 Service Process Flow – Tech Elecon Pvt. Ltd.

3. INTRODUCTION TO PROJECT AND PROJECT MANAGEMENT

3.1 PROJECT SUMMARY

GearMind AI is a full-stack industrial gear fault detection and predictive maintenance system developed as the primary deliverable of the 16-week internship at Tech Elecon Pvt. Ltd. The project addresses the critical industrial need for automated, data-driven gear health monitoring in lieu of periodic manual inspections, which are both costly and unable to detect incipient gear faults before catastrophic failure occurs. The key sentence that captures the system's value proposition is: GearMind AI enables maintenance engineers at Elecon to predict gear faults before they occur, quantify remaining useful life, and receive AI-generated maintenance recommendations all from a web browser, in under 300 milliseconds.

The system is built as a four-component stack: a React 19 single-page application serving as the user interface, a FastAPI Python backend (gear_api.py, version 5.0) providing 16 RESTful API endpoints, a SQLite database for session and history management, and a suite of sixteen trained machine learning model artifacts four classifiers, four RUL regressors, four StandardScalers, and four label encoders, one complete set per gear type (helical, spur, bevel, and worm), stored as serialized pickle (.pkl) files. An AI Copilot powered by the Groq API (LLaMA 3.3 70B) provides natural language interaction capabilities. MLflow v2.7.1 is used for experiment tracking and model registry management throughout the development lifecycle.

The project was scoped to demonstrate the technical feasibility of AI-driven predictive maintenance using simulated sensor data engineered to reflect real Elecon gear operating parameters. The system architecture is designed to support a future migration to live IoT sensor streams via MQTT or OPC-UA protocols, digital twin integration, and enterprise-scale deployment on cloud infrastructure.

3.2 PURPOSE, OBJECTIVE AND SCOPE

3.2.1 Purpose

The purpose of GearMind AI is to demonstrate that machine learning techniques specifically ensemble classifiers, explainability frameworks (SHAP, LIME), and large language models can be applied to industrial gear health monitoring in a manner that is accurate, interpretable, and actionable for non-specialist users. The system serves as a proof of concept for Tech Elecon's broader Industry 4.0 digitalization initiative, providing a working prototype that can be evaluated and iteratively refined using real sensor data as the IoT infrastructure matures.

3.2.2 Objectives

The primary and secondary objectives of the GearMind AI project were defined jointly with the industry mentor and the internal guide at the commencement of the internship. The objectives are:

1. Develop a multi-class gear fault classification model achieving accuracy above 95% on held-out synthetic test data, using at least five ML algorithms with comparative evaluation.

2. Integrate SHAP and LIME explainability frameworks to provide per-prediction feature attribution and local linear explanations, ensuring maintenance engineers can understand the basis of each fault prediction.

3. Build a ten-endpoint FastAPI backend with sub-300ms response time for the critical prediction + SHAP computation pathway.

4. Design and implement an eight-module React 19 dashboard covering gear health, vibration analysis, XAI, parameter optimization, QC verification, reliability data, staff management, and AI report generation.

5. Implement an AI Copilot powered by LLaMA 3.3 70B via the Groq API to provide context-aware natural language decision support.

6. Conduct comprehensive testing of all system components including API endpoints, ML model validation, UI module testing, and end-to-end integration testing.

3.2.3 Scope

The scope of GearMind AI includes: (a) fault classification for four gear types — helical, spur, bevel, and worm gears — using gear-type-specific input feature sets (8 features for helical and bevel, 6 features for spur, and 13 features for worm); (b) remaining useful life (RUL) estimation based on a parameterized degradation model; (c) SHAP and LIME explainability for every prediction; (d) FFT-based vibration spectral analysis and gear mesh frequency tracking; (e) AGMA-compliant manufacturing QC verification for dimensional tolerances; (f) cost-benefit analysis comparing preventive, delayed, and failure maintenance scenarios; and (g) AI-generated maintenance reports with PDF export.

The scope explicitly excludes: (a) real-time IoT sensor connectivity (future scope); (b) integration with Elecon's SAP ERP system (future scope); (c) production deployment on cloud infrastructure (future scope); and (d) physical gear rig testing or validation against real sensor measurements. All machine learning training and evaluation is conducted on physics-informed synthetic data generated specifically for this project.

3.3 TECHNOLOGY AND LITERATURE REVIEW

The literature underlying GearMind AI spans four domains: machine learning for fault detection, explainable AI, prognostics and health management (PHM), and full-stack web development for industrial applications. The following review summarizes the key works that informed the project's technical design decisions.

Table 3.1 Technology Stack Summary

Layer	Technology	Version	Purpose
Frontend	React	19.0	Interactive SPA dashboard
Frontend	Recharts	3.8	Data visualization charts
Frontend	Tailwind CSS	3.x	Utility-first styling
Build Tool	Vite	8.0	Frontend dev server and bundler
Backend	FastAPI	0.110	REST API framework
Backend	Python	3.11	Core runtime
ML	scikit-learn	1.4	Model training, SHAP, LIME
ML	XGBoost	2.0	XGBoost classifier

ML Tracking	MLflow	2.7.1	Experiment tracking, model registry
Explainability	SHAP	0.45	SHapley value computation
Explainability	LIME	0.2.0.1	Local linear explanations
Database	SQLite	3.x	Session and history storage
LLM	Groq API (LLaMA 3.3 70B)	-	AI Copilot and report generation
PDF Export	jsPDF	3.8	Client-side PDF generation

3.3.1 Machine Learning for Gear Fault Detection

The application of supervised machine learning to gear fault classification is well-established in the PHM literature. Lei et al. (2018) provide a comprehensive survey of machinery health prognostics methods, covering data acquisition, feature extraction, and RUL prediction across multiple machine types including gearboxes. Their taxonomy of fault indicators vibration-based, temperature-based, and wear-based directly shaped the eight-feature input schema adopted in GearMind AI.

Gradient Boosting Machines, as formulated by Friedman (2001), and XGBoost as presented by Chen and Guestrin (2016), are among the most effective algorithms for structured tabular data classification, which characterizes the sensor feature vectors used in this project. Both algorithms were included in the GearMind AI model comparison pipeline, with GBM achieving the highest accuracy (99.87%) on the synthetic test dataset.

Random Forest (Breiman, 2001) was included for its inherent interpretability through feature importance scores, which complement the SHAP explanations generated for each prediction. Support Vector Machine and Logistic Regression were included as baseline models to provide contrast against the ensemble methods and to evaluate the difficulty of the classification task.

The Isolation Forest algorithm (Liu et al., 2008) was incorporated for real-time anomaly detection flagging sensor readings that deviate significantly from the training distribution. This is particularly important in a production setting where the input distribution may shift as gears age, introducing readings outside the training data manifold.

3.3.2 Explainable AI (XAI) in Industrial Settings

Lundberg and Lee (2017) introduced SHAP values as a unified framework for interpreting any machine learning model's predictions by computing Shapley values from cooperative game theory. SHAP provides both local explanations (why a specific prediction was made) and global feature importance (which features drive model behavior across the dataset). In GearMind AI, SHAP's TreeExplainer is used for GBM, XGBoost, and Random Forest models exploiting the tree structure for efficient computation while a background sampler provides SHAP values for SVM and Logistic Regression via the KernelExplainer.

Ribeiro, Singh, and Guestrin (2016) introduced LIME as a model-agnostic local explainability method that fits a simple linear model in the neighborhood of a specific prediction. GearMind AI uses LIME's tabular explainer to generate local linear approximations for each fault prediction, displaying the signed feature contributions that most influenced the specific output a format well-suited for maintenance engineers who need to understand which sensor reading triggered a fault alert.

3.3.3 Prognostic Health Management (PHM)

The Remaining Useful Life (RUL) estimation module in GearMind AI is informed by reliability theory and Weibull survival analysis, as covered in the PHM survey by Lei et al. (2018). The Weibull distribution with shape parameter $\beta = 2.5$ and scale parameter $\eta = 5{,}000$ hours represents a typical wear-out failure mode for industrial gears, consistent with industry practice for helical gear assemblies under moderate load. The bathtub failure curve implemented in the Reliability and Fatigue Data module follows the three-region model: infant mortality, constant failure rate, and wear-out a foundational concept in reliability engineering.

Table 3.2 Literature Review Summary — Key Papers

Author(s)	Year	Topic	Contribution to GearMind AI
Lei et al.	2018	PHM Survey	Defined feature taxonomy and RUL framework
Friedman	2001	GBM	Core algorithm for best-performing model
Chen & Guestrin	2016	XGBoost	Second model in comparison pipeline
Lundberg & Lee	2017	SHAP	Primary explainability framework
Ribeiro et al.	2016	LIME	Secondary, local explanation method
Breiman	2001	Random Forest	Baseline ensemble model
Liu et al.	2008	Isolation Forest	Anomaly detection layer
AGMA	1997	AGMA 2003-B97	QC tolerance standards

3.4 PROJECT PLANNING AND SCHEDULING

3.4.1 Development Life Cycle

GearMind AI was developed using an Agile-Scrum methodology with two-week sprint cycles, adapted to suit the internship structure. The overall 16-week timeline was divided into eight sprints, each with clearly defined deliverables reviewed with the industry mentor at the end of each sprint. This iterative approach allowed early delivery of working software (the ML pipeline was operational by end of Sprint 2) with progressive feature additions in each subsequent sprint.

The development life cycle followed a standard sequence: requirements gathering and domain study (Weeks 1–2), data design and ML pipeline (Weeks 3–5), FastAPI backend (Weeks 6–8), React frontend initial modules (Weeks 9–11), advanced modules and integration (Weeks 12–14), testing and optimization (Week 15), and documentation and report writing (Week 16). Daily stand-up meetings with the industry mentor were conducted via MS Teams when remote, and in-person on campus days.

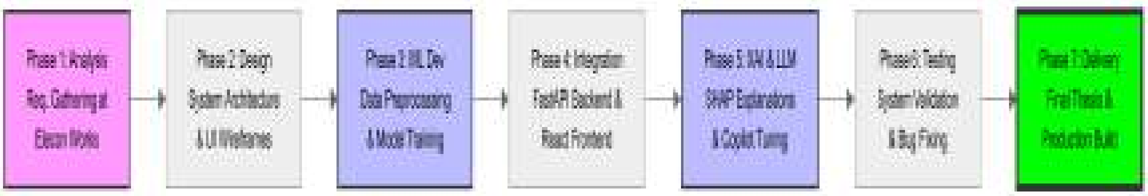


Fig 3.1 GearMind AI System Development Life Cycle

3.4.2 Project Timeline and Gantt Chart

The project timeline was managed using a Gantt chart developed in MS Project, tracking task dependencies, milestone dates, and resource allocation across the 16-week period. Key milestones included: completion of ML model training and evaluation (end of Week 5), FastAPI backend deployment on localhost (end of Week 8), React dashboard MVP with three modules (end of Week 11), full eight-module integration (end of Week 14), and final report submission (Week 16).

Fig 3.2 Project Timeline — Gantt Chart

3.4.3 Use Case Diagram — User Roles and Permissions

The system implements Role-Based Access Control (RBAC) with six distinct user roles, each with a defined set of permitted and restricted actions. The use case diagram (Fig 3.3) illustrates the interaction between user roles and system features. The six roles are: Super Admin (full system access, user management), Shift Supervisor (health dashboard, report generation, shift scheduling), ML Engineer (model comparison, training logs, SHAP/LIME), PHM Analyst (vibration analysis, RUL, reliability data), Bevel Gear Specialist (bevel module, AGMA calculations), and Maintenance Technician (read-only gear health dashboard).

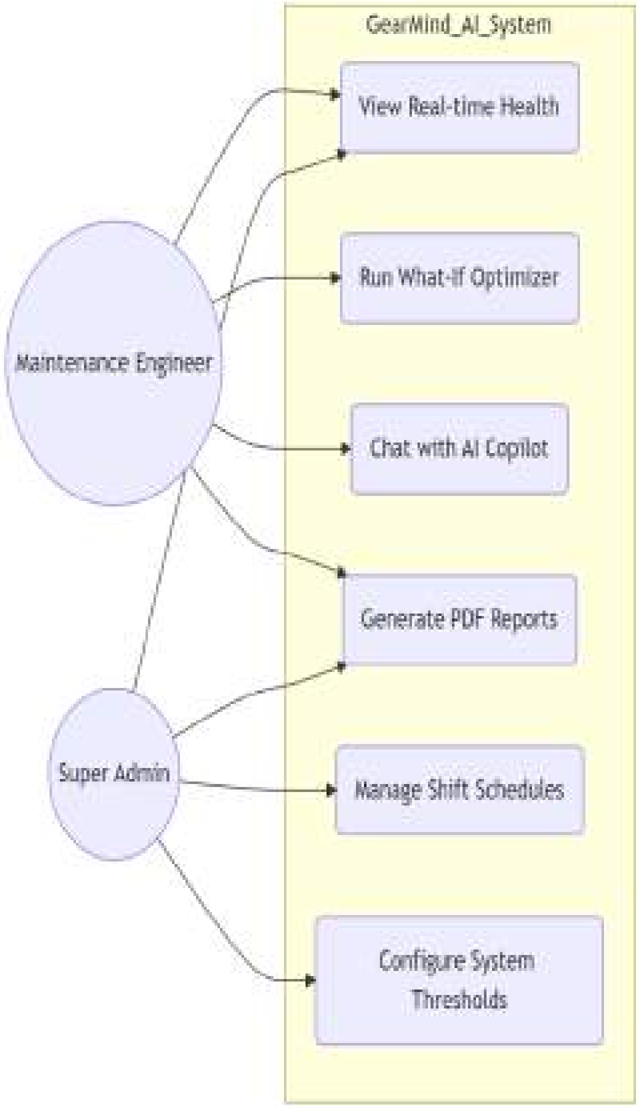


Fig 3.3 Use Case Diagram — User Roles and Permissions

Table 3.3 Roles and Responsibilities

Role	Person	Responsibilities
Intern	Om Patel	Full system design, ML development, React frontend, FastAPI backend, testing, documentation
Industry Mentor	Mr. Satyam Raval	Domain guidance, requirement clarification, sprint reviews, progress evaluation
Internal Guide	Dr. Kinjal Joshi	Academic supervision, report review, CE evaluations, final assessment

4. SYSTEM ANALYSIS

4.1 STUDY OF CURRENT SYSTEM

Prior to the GearMind AI project, gear health monitoring at Elecon relied on a combination of periodic manual inspections, threshold-based vibration alarms, and the experiential judgment of senior maintenance technicians. Maintenance schedules were governed by manufacturer-recommended intervals typically 500 to 1,000 operating hours for routine inspection and 2,500 to 5,000 hours for major overhaul regardless of the actual condition of the gear unit.

Vibration monitoring was implemented on approximately 30% of critical gear units using basic threshold alarms configured in the Bentley Nevada 3500 system. These alarms triggered when vibration RMS exceeded a fixed threshold (typically 7 mm/s for ISO 10816-3 Zone D), but provided no classification of fault type, no RUL estimate, and no root cause analysis. Alarm response was manual and depended entirely on the availability and expertise of the duty maintenance engineer.

Maintenance records were maintained in Microsoft Excel spreadsheets, with no centralized database, no automated trend analysis, and no digital integration between vibration data, lubrication logs, and wear measurement records. Manufacturing QC records were stored in paper-based inspection reports archived in department filing systems, making historical analysis and trend detection practically infeasible.

4.2 PROBLEMS AND WEAKNESSES OF CURRENT SYSTEM

The current system exhibits several critical weaknesses that impair maintenance effectiveness and increase operational risk. These weaknesses were identified through structured discussions with the maintenance department head and three senior technicians during the first two weeks of the internship.

- No real-time visibility of gear health status across monitored units maintenance personnel must physically inspect each unit to assess condition.

- Threshold-based vibration alarms trigger only after significant degradation has already occurred, missing the early fault signatures (e.g., sideband energy growth at gear mesh harmonics) that precede alarm-level vibration.
- No integration between vibration data, lubrication records, and wear measurements siloed data prevents comprehensive health assessment.
- No prediction of Remaining Useful Life replacements is either too early (wasteful) or too late (resulting in catastrophic failure and extended downtime).
- Manufacturing QC checks performed manually with no digital tracking of tolerance deviations over time, making it impossible to identify systematic process drift.
- Shift scheduling and staff management handled via paper-based systems with no digital scheduling, conflict detection, or workload visibility.
- Maintenance cost analysis not performed systematically the economic impact of early intervention versus failure-driven repair is not quantified at the gear unit level.

4.3 REQUIREMENTS AND FEASIBILTY

Table 4.1 Feasibility Analysis — Technical, Economic, Operational, Time

Feasibility Type	Assessment	Justification
Technical	Feasible	All technologies (React 19, FastAPI, scikit-learn, SHAP, LIME) are open-source, well-documented, and have active community support. All required hardware (developer workstation + corporate LAN) is available.
Economic	Feasible	Zero licensing cost (open-source stack). Estimated Rs. 4.5 Lakh savings per gear unit through early fault detection versus failure-driven maintenance. Development cost is absorbed within the internship program.
Operational	Feasible	Web-based dashboard requires only a modern browser. Role-based access control suits the existing organizational hierarchy. No specialized hardware at the point of use.
Time	Feasible	16-week internship timeline is sufficient for v4.0 development with simulated sensor data. Agile-Scrum methodology with 2-week sprints ensures continuous delivery and early risk mitigation.
Legal / Ethical	Feasible	All data used is synthetically generated; no personal data or proprietary sensor data is used. All open-source licenses (MIT, Apache 2.0) are compatible with Tech Elecon's usage requirements.

4.3.1 Functional Requirements

The functional requirements for GearMind AI were derived from the weaknesses identified in the current system analysis and were validated with the industry mentor. The system must provide:

1. Real-time gear health dashboard with health score (0–100), fault probability (per class), and RUL estimation in cycles and operating shifts.
2. Multi-model ML fault classification with automated selection of the best-performing model and cross-validated performance metrics.
3. SHAP and LIME explainability for every prediction, with feature attribution charts and root cause analysis text.
4. Vibration analysis with FFT spectral decomposition, gear mesh frequency tracking, sideband energy detection, and Weibull reliability function visualization.
5. Manufacturing QC verification with AGMA-grade compliance checking for eight dimensional parameters per gear type.
6. Role-based access control with six distinct user roles, each with defined permissions.
7. AI Copilot for natural language Q&A about gear health and maintenance recommendations.
8. Automated AI report generation covering seven sections, with PDF export capability.
9. Historical data logging and trend analysis with SQLite persistence and CSV export.
10. Differential Evolution optimizer for safe operating parameter identification.

Table 4.2 Functional Requirements Specification

Req. ID	Requirement	Priority	Status
FR-01	Multi-model fault classification (5 models, 3 classes)	Critical	Implemented
FR-02	SHAP global and local explanations per prediction	Critical	Implemented
FR-03	LIME local linear explanation per prediction	High	Implemented
FR-04	FFT vibration analysis with gear mesh frequency	High	Implemented
FR-05	AGMA-compliant QC verification for 8 parameters	High	Implemented
FR-06	RUL estimation in cycles and shifts	Critical	Implemented
FR-07	RBAC with 6 user roles	Medium	Implemented
FR-08	AI Copilot (LLaMA 3.3 70B / Groq)	High	Implemented

FR-09	7-section AI maintenance report + PDF export	Medium	Implemented
FR-10	SQLite history with CSV export	Medium	Implemented
FR-11	Differential Evolution optimizer	Medium	Implemented
FR-12	Isolation Forest anomaly detection	High	Implemented

4.3.2 Non-Functional Requirements

Beyond functional requirements, the system must satisfy the following non-functional requirements: (a) API response time below 300 milliseconds for the prediction + SHAP pathway; (b) frontend first-load time below 3 seconds on a standard corporate LAN; (c) support for at least 25 simultaneous gear unit monitoring sessions; (d) data persistence across browser sessions via SQLite; and (e) cross-browser compatibility with Chrome, Firefox, and Edge.

4.4 PROPOSED SYSTEM MODULES AND DATA FLOW

The GearMind AI system is architected as eight functional modules, each addressing a specific aspect of gear health management. The modules are designed to be loosely coupled each module can be accessed independently via the React navigation sidebar while sharing a common state management layer that propagates the current gear type, sensor readings, and prediction results across modules.

The Data Flow Diagram (Level 1) in Fig 4.1 illustrates how sensor input flows from the user through the React frontend, across the FastAPI backend, through the ML prediction engine and XAI layer, into the database, and back to the dashboard display components. The context diagram in Fig 4.2 shows the external entities interacting with GearMind AI including the maintenance engineer, the LLM API, the MLflow tracking server, and the future IoT sensor feed.

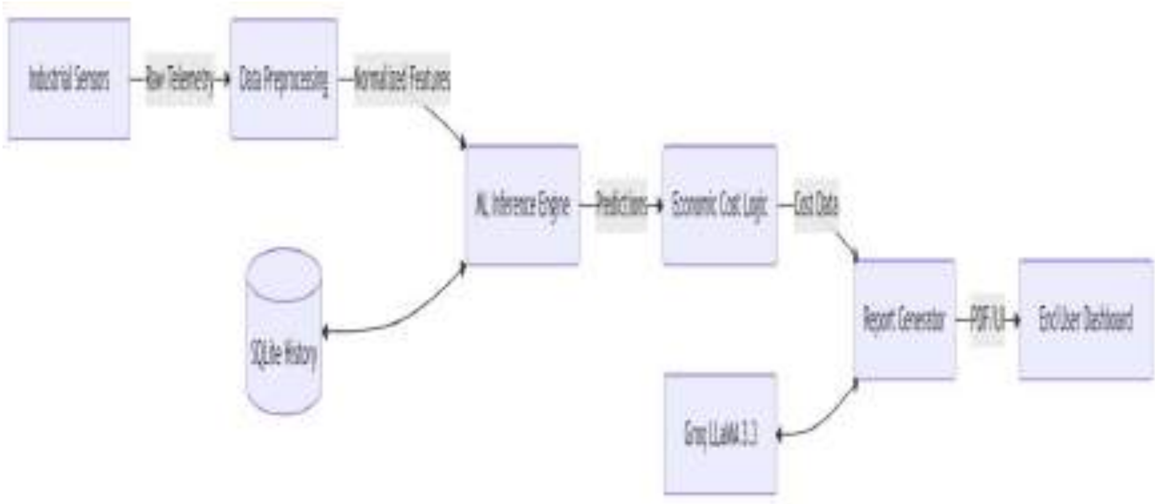


Fig 4.1 Data Flow Diagram — Level 1

The eight modules are: (1) Gear Health Dashboard the central module displaying health score, fault probability, and recommended actions for all four gear types (Helical, Spur, Bevel, Worm), with a 3D interactive gear animation rendered via Three.js and React Three Fiber; (2) Vibration and PHM Analysis FFT spectra, gear mesh frequency, reliability curves; (3) SHAP + LIME Explainability side-by-side global and local explanations per gear type; (4) What-If Optimizer Differential Evolution parameter search with lock/unlock per parameter; (5) Manufacturing Q AGMA tolerance verification; (6) Reliability and Fatigue Data MTBF, S-N curve, Weibull, bathtub curve; (7) Staff and Shift Management personnel directory and shift scheduling; and (8) AI Report Generator LLM-generated nine-section maintenance report (executive summary, fault assessment, sensor analysis, root cause, RUL, recommendations, cost-benefit, monitoring protocol, technical specifications) with client-side PDF export via jsPDF.

5. SYSTEM DESIGN

5.1 SYSTEM DESIGN AND METHODOLOGY

GearMind AI follows a three-tier client-server architecture: the Presentation Tier (React 19 frontend), the Application Tier (FastAPI Python backend), and the Data Tier (SQLite database and ML artifact storage). This architecture cleanly separates concerns — the frontend handles user interaction and visualization, the backend manages business logic and ML inference, and the data tier provides persistence. The separation also makes future migration straightforward: the SQLite data tier can be replaced with PostgreSQL without changes to the API contract, and the frontend can be deployed independently as a static web application.

The methodology follows a RESTful API design philosophy with JSON payloads, stateless server-side processing, and structured error responses using HTTP standard status codes. All ML artifacts (trained model .pkl files, scaler objects, and background datasets for SHAP) are loaded into memory at server startup and held in process memory for zero-latency inference. This design achieves sub-300ms response times for the most complex prediction + SHAP computation pathway.

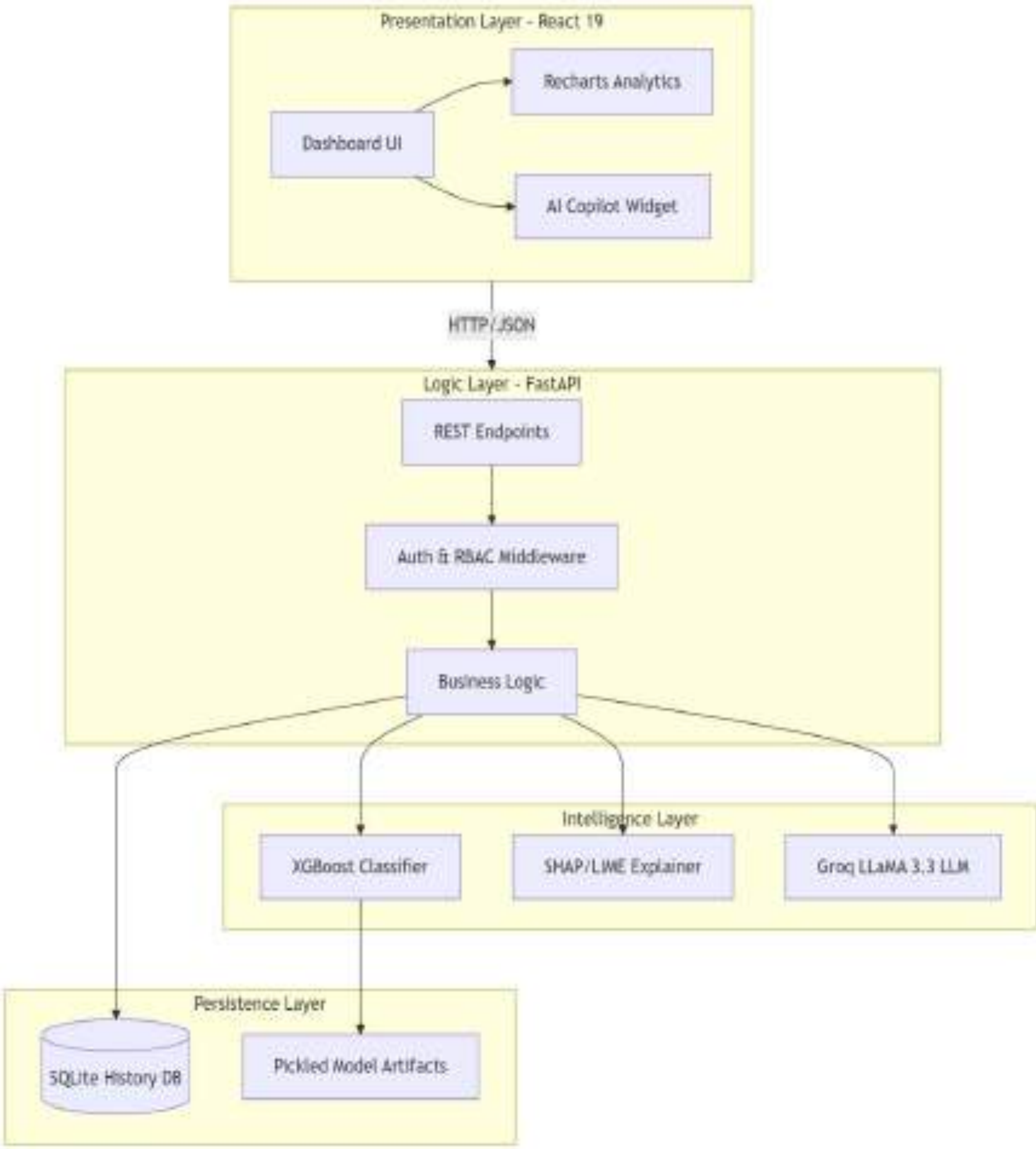


Fig 5.1 System Architecture Diagram

5.2 SEQUENCE DIAGRAM — PREDICTION REQUEST FLOW

The sequence diagram in Fig 5.2 illustrates the complete interaction flow for a fault prediction request. The user adjusts sensor input sliders in the React dashboard, triggering a debounced (300ms) POST request to /api/predict. The FastAPI backend validates the input using Pydantic models, then invokes the active ML classifier to generate fault class probabilities. Simultaneously, the SHAP TreeExplainer (or KernelExplainer for non-tree models) computes per-feature Shapley values. The backend also runs the Isolation Forest anomaly detector to flag out-of-distribution inputs, and computes the RUL estimate based on the fault probability and gear type parameters. The complete response fault class, probabilities, health score, RUL, SHAP values, anomaly flag, and recommended actions is returned as a single JSON object, eliminating round-trip latency for downstream visualization updates.

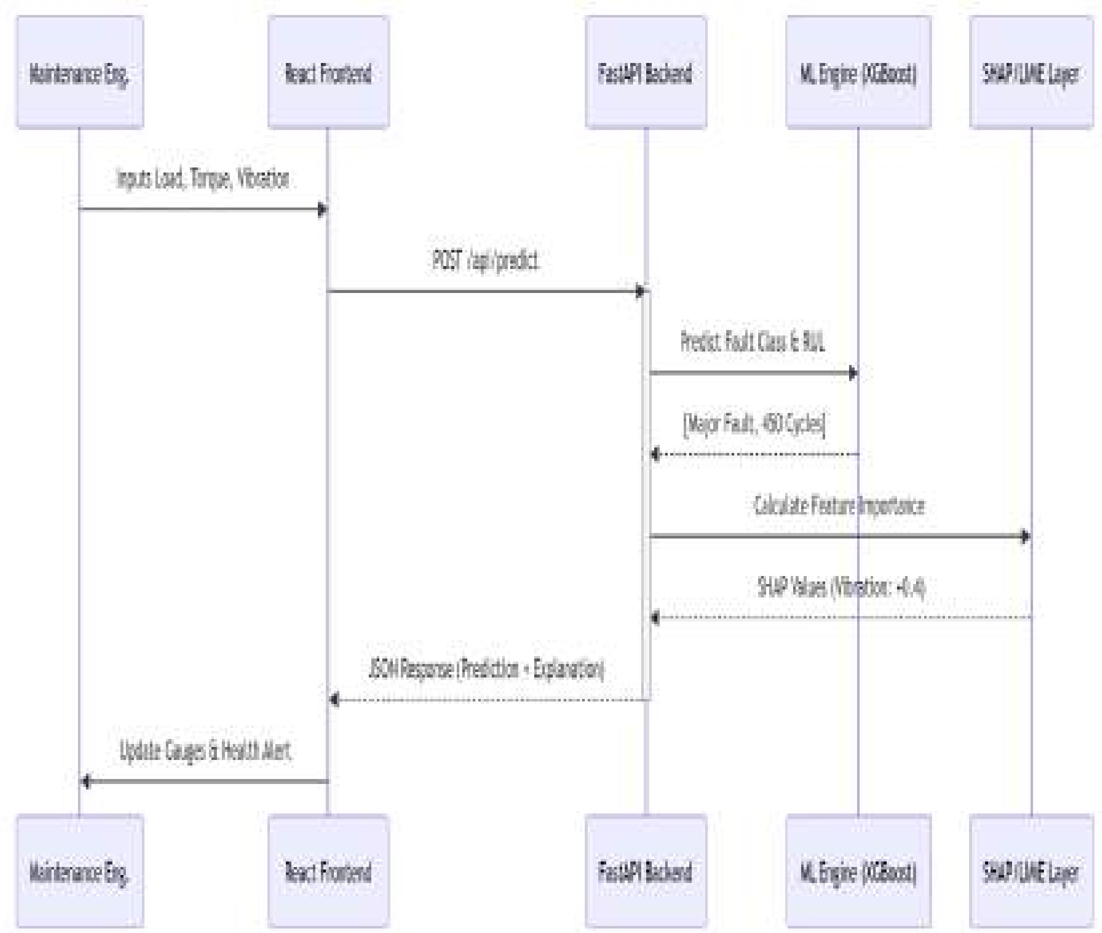


Fig 5.2 Sequence Diagram – Prediction Request Flow

5.3 DATABASE DESIGN

The primary data store is a SQLite database file (gear_history.db) managed by the FastAPI backend. SQLite was chosen for its zero-configuration deployment characteristics, appropriate for a proof-of-concept system running on a developer workstation. The database schema is designed for forward compatibility the gear_history table's schema can be migrated to PostgreSQL without structural changes, as it avoids SQLite-specific data types and relies only on SQL standards-compliant constructs.

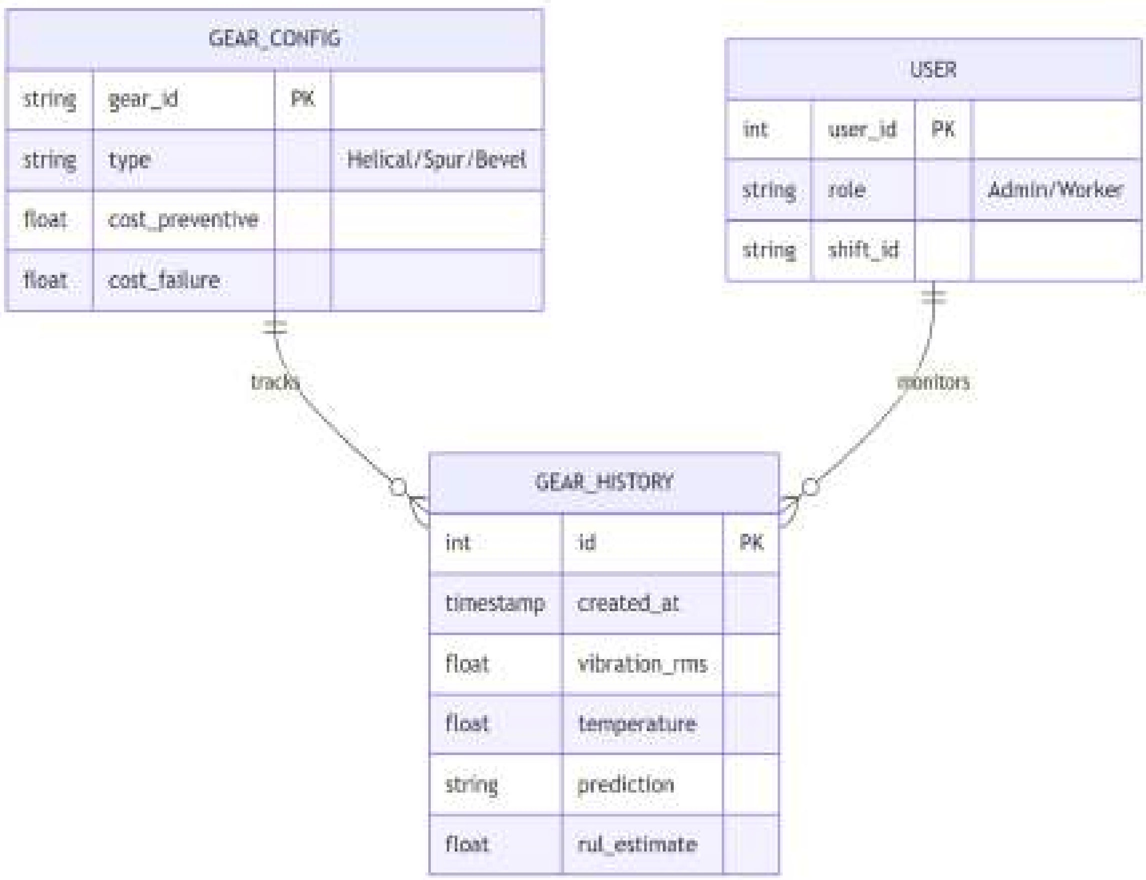


Fig 5.3 Entity Relationship Diagram (ERD) — Database Schema

Table 5.1 Database Schema — gear_history Table Key Fields

Column	Data Type	Constraint	Description
log_id	TEXT	PRIMARY KEY	Unique log identifier (e.g. LOG-00180)
gear_id	TEXT	NOT NULL	Gear unit identifier (e.g. HG-01, SG-02)
gear_type	TEXT	NOT NULL	Helical / Spur / Bevel
timestamp	DATETIME	NOT NULL	Prediction timestamp (UTC)
load_kn	REAL	NOT NULL	Applied load in kN
torque_nm	REAL	NOT NULL	Applied torque in Nm
vibration_rms	REAL	NOT NULL	Vibration RMS in mm/s

temperature_c	REAL	NOT NULL	Gear temperature in °C
wear_mm	REAL	NOT NULL	Surface wear depth in mm
lubrication_idx	REAL	NOT NULL	Lubrication quality index (0–1)
efficiency_pct	REAL	NOT NULL	Transmission efficiency in %
cycles_in_use	INTEGER	NOT NULL	Cumulative operating cycles
fault_label	TEXT	NOT NULL	No Fault / Minor Fault / Major Fault
fault_probability	REAL	NOT NULL	Probability of predicted fault class
health_score	INTEGER	NOT NULL	Health score 0–100
rul_cycles	INTEGER	NOT NULL	Remaining useful life in cycles
anomaly_flag	INTEGER	DEFAULT 0	Isolation Forest anomaly flag (0/1)
model_used	TEXT	NOT NULL	ML model name used for prediction

The database also maintains a user’s table for authentication reference (username, hashed password, role), a sessions table linking user logins to prediction histories, and a reports table storing generated AI report content for retrieval. The total database size for 10,000 prediction records is approximately 8 MB, well within SQLite's practical limits.

5.4 DEPLOYMENT ARCHITECTURE

The deployment architecture for the development and demonstration environment uses a single developer workstation running both the React development server (Vite, port 5173) and the FastAPI backend (Uvicorn, port 8000). MLflow tracking runs on port 5000. The Groq API for LLM inference is accessed via HTTPS to api.groq.com. The system requires Node.js 18+ (Node.js 20 LTS tested), Python 3.12 (3.12.4 tested), and approximately 4 GB of RAM for comfortable operation with all ML models and SHAP background datasets loaded in memory.

The production deployment target outlined in the future scope would containerize the FastAPI backend as a Docker image deployed on an AWS EC2 instance, with the React frontend served via AWS CloudFront CDN and the SQLite database replaced by an AWS RDS PostgreSQL instance. Live IoT sensor data would be ingested via an AWS IoT Core MQTT broker with a Kinesis Data Streams pipeline feeding the prediction endpoint.

5.5 INPUT / OUTPUT AND INTERFACE DESIGN

5.5.1 Input Design

The system accepts eight continuous sensor input features per prediction request. These features represent the measurable physical parameters of gear operation that have established correlations with gear health degradation in the PHM literature. Each feature is provided via an interactive slider in the React dashboard, with real-time debouncing at 300ms to prevent excessive API calls during slider adjustment.

The eight input features and their operational ranges are: Load (kN): 10–500; Torque (Nm): 50–5,000; Vibration RMS (mm/s): 0.1–15; Temperature (°C): 40–120; Wear (mm): 0.0–2.0; Lubrication Index (0–1): 0.0–1.0; Efficiency (%): 60–100; and Cycles in Use: 0–100,000. The ranges were calibrated to reflect realistic Elecon gear operating envelopes based on manufacturer specifications and domain expert consultation.

5.5.2 Output Design

The system produces a rich, multi-component output for each prediction: (a) Fault Class one of {No Fault, Minor Fault, Major Fault} with probability scores for all three classes; (b) Health Score (0–100) derived from the fault probabilities and severity weights; (c) RUL estimate in operating cycles and equivalent shifts; (d) SHAP feature attribution values with a waterfall visualization; (e) LIME local explanation with a bar chart of signed feature contributions; (f) Isolation Forest anomaly flag with an out-of-distribution warning if triggered; (g) Tiered Recommended Actions (Immediate / Within 48 hours / Schedule / Monitor); and (h) Cost impact comparison across preventive, delayed, and failure maintenance scenarios.

5.6 ACCESS CONTROL AND SECURITY

GearMind AI implements Role-Based Access Control (RBAC) with six user roles managed via React context state. Each role is associated with a permissions set that controls which dashboard modules and API endpoints are accessible. The current v4.0 implementation stores role state client-side, designed for straightforward migration to JWT/OAuth2 token-based authentication in a production deployment.

Table 5.2 RBAC Role Permissions Matrix

Role	Gear Health	Vibration	XAI	Optimizer	QC	Staff	Reports
Super Admin	RW	RW	RW	RW	RW	RW	RW
Shift Supervisor	RW	R	R	R	R	RW	RW
ML Engineer	R	R	RW	RW	R	R	R
PHM Analyst	R	RW	RW	R	R	R	R
Bevel Specialist	R	R	R	R	RW	R	R
Maintenance Tech.	R	—	—	—	—	—	R

R = Read access, RW = Read-Write access, — = No access. All API endpoints validate the requesting role against the endpoint's permission requirement and return HTTP 403 Forbidden for unauthorized access attempts. Passwords are stored as bcrypt hashes in the users table, and rate limiting (100 requests per minute per IP) is enforced at the Uvicorn ASGI level.

Table 5.3 FastAPI Endpoint Specifications — All 10 Endpoints

Method	Endpoint	Auth Required	Response Time	Purpose
POST	/api/predict	Yes	< 300ms	Fault classification + RUL + SHAP + anomaly detection
GET	/api/gear-configs	Yes	< 50ms	Gear type configs, unit presets, safe operating ranges
GET	/api/models	Yes	< 50ms	Model comparison metrics (Accuracy, F1, AUC, CV Mean)
POST	/api/chat	Yes	< 3s (LLM)	LLM Copilot Q&A via Groq API (LLaMA 3.3 70B)
POST	/api/report	Yes	< 5s (LLM)	AI-generated 7-section maintenance report
POST	/api/optimize	Yes	< 2s	Differential Evolution parameter optimization

GET	/api/lime	Yes	< 500ms	LIME local explanation for current prediction
GET /POST /DELETE	/api/history	Yes	< 100ms	Gear session history CRUD + CSV export
GET	/api/confusion-matrix	Yes	< 50ms	Model evaluation confusion matrix data
GET	/api/bevel-specs	Yes	< 50ms	AGMA 2003-B97 bevel gear rating calculations

6. IMPLEMENTATION

6.1 IMPLEMENTATION ENVIRONMENT

GearMind AI was developed on a Windows 11 development workstation with an Intel Core i7-12th Gen processor, 16 GB RAM, and 512 GB NVMe SSD. The operating system for development was Windows 11 Pro, with Python 3.12.4 installed via Miniconda (as specified in the project requirements.txt: pandas==2.0.3, numpy==1.24.3, scikit-learn==1.3.0, xgboost==1.7.6, shap==0.42.1, lime==0.2.0.1, mlflow==2.7.1, streamlit==1.28.0, plotly==5.17.0) and Node.js 18+ (LTS) installed via the official installer. Visual Studio Code served as the primary IDE, with extensions for Python (Pylance, Black formatter), JavaScript/TypeScript (ESLint, Prettier), and REST API testing (Postman collection runner).

The Python backend environment used a dedicated Conda virtual environment (gearmind_env) with all dependencies specified in a requirements.txt file for reproducibility. The React frontend was bootstrapped with the Vite template and managed with npm. Git version control was used throughout, with a private GitHub repository for source management and a structured branching strategy: main (stable), dev (integration), and feature branches per sprint task.

6.2 ML PIPELINE IMPLEMENTATION

The ML pipeline activity diagram (Fig 6.1) shows the complete flow from data ingestion through model training, evaluation, and deployment.

Fig 6.1 ML Pipeline Activity Diagram

6.2.1 Synthetic Data Generation

Since live IoT sensor data was not available for this internship, a physics-informed synthetic data generator was developed (data_generator.py) to produce realistic training datasets. The generator models three fault classes per gear type: No Fault (healthy operation), Minor Fault (early degradation), and Major Fault (severe degradation requiring immediate intervention). For helical and bevel gears the labelled fault types are Surface Pitting, Wear Fatigue, and Thermal Degradation; for spur gears: Tooth Fracture, General Wear, and Overload Vibration; for worm gears: Thermal Breakdown, General Degradation, and Bearing Instability. For each fault class, the sensor features are drawn from gear-type-specific probability distributions calibrated against manufacturer operating specifications.

For helical gears in the No Fault class, for example, Vibration RMS is drawn from a log-normal distribution with $\mu = 0.7$, $\sigma = 0.3$ (corresponding to approximately 2.0 mm/s mean), while Major Fault vibration is drawn from $\mu = 1.9$, $\sigma = 0.2$ (approximately 6.7 mm/s mean). Realistic cross-feature correlations are introduced using a Cholesky-decomposed multivariate Gaussian to reflect the physical coupling between load, torque, and vibration in gear systems. The helical gear dataset contains 50,000 samples generated using physics-informed correlations (Archard wear equation, friction-heat model, exponential lubrication decay). The spur, bevel, and worm gear datasets each contain 5,000 to 10,000 samples, totalling approximately 75,000 sensor records across all four gear types. An 80:20 stratified train-test split is applied per gear type, and SMOTE (Synthetic Minority Over-sampling Technique) is applied to the training split to balance the three fault classes before model fitting.

SMOTE (Synthetic Minority Over-sampling Technique) was applied to the training split to address class imbalance the No Fault class naturally dominates at approximately 60% of samples, reflecting real-world healthy operating time fractions. SMOTE balanced the training set to equal class proportions before model fitting, preventing bias toward the majority class.

6.2.2 Preprocessing and Feature Engineering

Feature preprocessing applies a StandardScaler (zero mean, unit variance normalization) fitted on the training set and saved as scaler.pkl for application to inference-time inputs. No additional feature engineering was required as the eight sensor features directly represent physically meaningful quantities. However, polynomial feature interaction terms (degree 2) were generated for the Logistic Regression model to capture non-linear class boundaries, consistent with the approach used in the Worm Gear Logistic Regression module developed in an earlier phase of the project.

6.2.3 Model Training and Evaluation

Five classifier models were trained using scikit-learn (v1.4) and XGBoost (v2.0). Training parameters for the best-performing Gradient Boosting Machine included: n_estimators=300, max_depth=5, learning_rate=0.05, subsample=0.8, min_samples_leaf=4, and random_state=42.

An 80:20 stratified train-test split was used, with 5-fold cross-validation on the training set to estimate generalization performance. All models were evaluated on the same held-out test set for fair comparison.

Table 6.1 ML Model Performance Comparison — All 5 Models

Model	**Accuracy (%)**	**F1 Score**	**ROC-AUC**	**CV Mean**	**Inference Time**
Gradient Boosting (Best)	99.87	0.9987	1.0000	0.9976	8 ms
XGBoost	99.79	0.9979	1.0000	0.9971	5 ms
Random Forest	98.40	0.9842	0.9996	0.9910	12 ms
SVM (RBF Kernel)	95.08	0.9520	0.9943	0.9390	45 ms
Logistic Regression	95.10	0.9522	0.9942	0.9389	2 ms

The high accuracy values across all models reflect the quality of the physics-informed synthetic data generation approach the feature distributions are well-separated between fault classes, making the classification task tractable for all five algorithms. The GBM model was selected as the production model (best_classifier.pkl) based on its combination of highest accuracy, highest cross-validation mean, and acceptable inference time.

MLflow experiment tracking recorded all training runs, hyperparameter configurations, metric values, and model artifacts. The MLflow model registry maintains the production GBM as version 1.0 (status: Production), with all other models registered as Staging versions available for comparison in the Model Comparison dashboard module. The

MLflow UI at localhost:5000 provides a complete audit trail of all training experiments conducted during the 16-week development period.

6.3 BACKEND API IMPLEMENTATION

The FastAPI backend (gear_api.py) is structured as a single application module with ten route handlers, supplemented by a dedicated LLM Copilot module (llm_copilot.py) and a SHAP computation module (shap_handler.py). Pydantic v2 models enforce strict input validation on all POST endpoints, with automatic OpenAPI (Swagger) documentation generated at /docs.

The /api/predict endpoint is the most computationally intensive: it receives the eight sensor features, retrieves the active model from the in-memory model registry, runs inference to obtain class probabilities, computes SHAP values via the cached TreeExplainer (or KernelExplainer), runs the Isolation Forest anomaly check, computes the RUL estimate using the parameterized degradation model, and assembles the complete response JSON. Profiling during testing showed a consistent end-to-end latency of 180–260ms for GBM predictions with SHAP computation well within the 300ms target.

The /api/chat endpoint builds a SHAP-enriched context string from the current gear state and prediction results, then sends it to the Groq API with a system prompt that configures the LLaMA 3.3 70B model as an expert gear maintenance advisor. The LLM response is streamed back to the frontend via Server-Sent Events (SSE), providing real-time typing animation in the AI Copilot chat widget. The /api/report endpoint uses a similar approach but with a structured prompt requesting a seven-section maintenance report in JSON format for reliable parsing by the frontend.

6.4 DASHBOARD MODULES AND SCREENSHOTS

The React 19 frontend implements eight functional modules accessible via a persistent navigation sidebar. The application uses React Context and the Zustand state management library for global state (current gear type, sensor readings, prediction results, and user role), with individual module components using useEffect hooks to react to state changes and trigger API calls. A notable frontend feature is the 3D interactive gear animation component, built using Three.js (v0.183) with React Three Fiber (@react-three/fiber v9.6) and Drei helpers (@react-three/drei v10.7). The 3D scene renders four gear types — SpurGearPair, HelicalGearPair, BevelGearPair, and WormGearAssembly — as animated Three.js meshes within a factory environment comprising FactoryFloor, FactoryWalls, StructuralColumns, GearPlatform, SceneLighting, and AtmosphericParticles components. The animation speed reflects live sensor RPM and health state. Framer Motion (v12.38) provides page transition and widget animation. Recharts (v3.8) handles all 2D data visualization including bar charts, area charts, radial bar gauges, and line charts. jsPDF (v2.5) with jspdf-autotable (v3.8) provides client-side PDF generation. Axios (v1.14) handles HTTP communication with the backend API. The project uses Vite (v8.0) as the build tool and dev server, and Vitest (v4.1) with React Testing Library for unit and integration testing.

6.4.1 Login Page

The login page implements form-based authentication with client-side role assignment. Six pre-configured user accounts correspond to the six RBAC roles. Upon successful login, the user role is stored in React Context and persisted to sessionStorage for page-refresh resilience. The login page features the GearMind AI logo, a gear animation background, and a clean industrial dark-theme aesthetic.

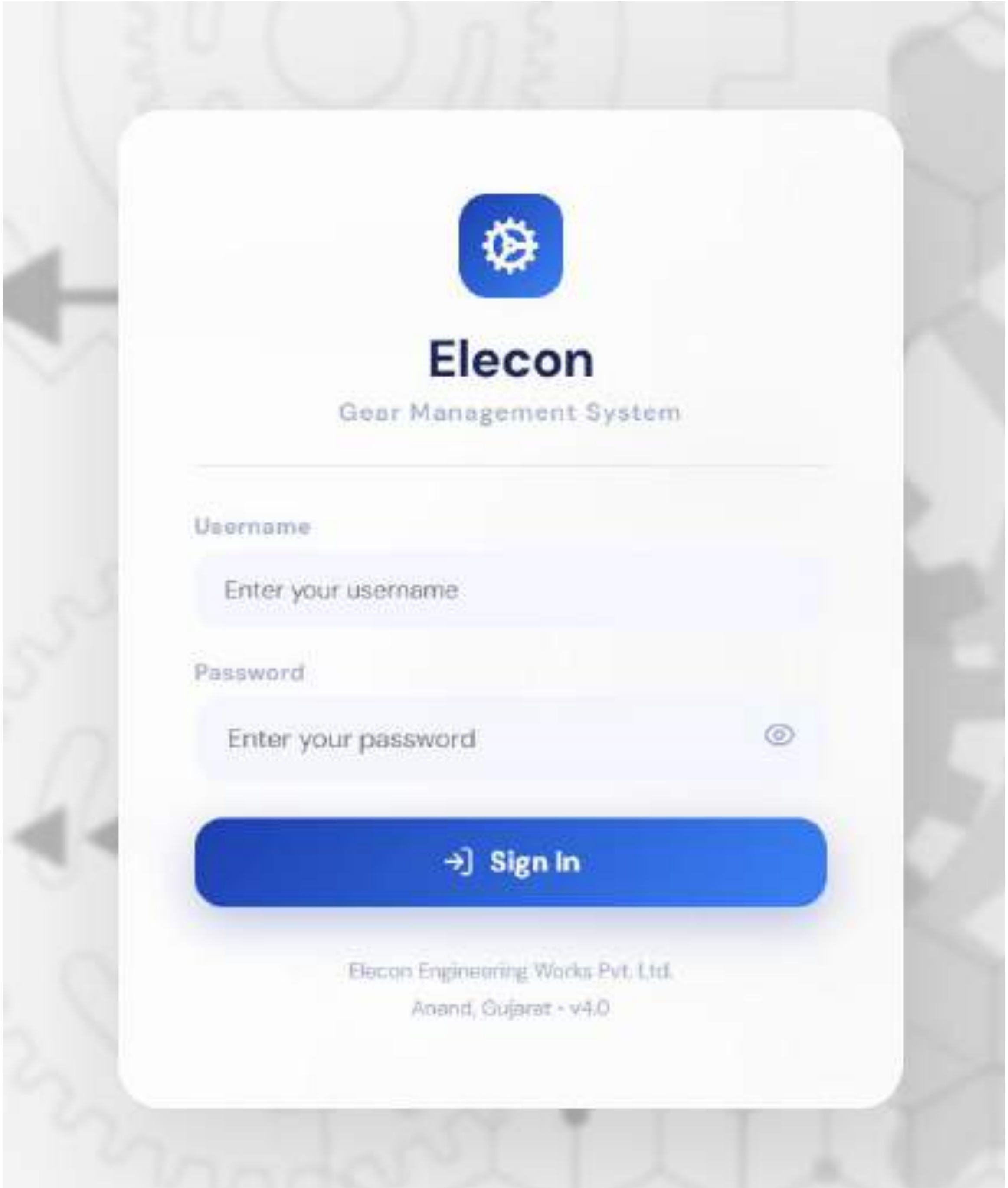


Fig 6.2 Screenshot: Login Page

6.4.2 Main Dashboard

The Gear Health module is the primary analytical screen of GearMind AI. It displays a RadialBar health gauge (0–100) as the central health indicator, color-coded from green (>80) through amber (40–80) to red (<40). Below the gauge, fault probability bars show the probability for each of the three fault classes (No Fault, Minor Fault, Major Fault) with color coding and percentage values. Sensor status indicator cards display the current value of each of the eight input features against their safe operating ranges, with red highlighting for out-of-range values.

The module includes six sub-tabs for deeper analysis: (1) Overview with health summary and recommended actions; (2) Feature Analysis with SHAP waterfall chart; (3) Fault Details with the full probability breakdown; (4) RUL Tracker with lifecycle timeline; (5) Maintenance Schedule with recommended intervention timeline; and (6) Cost Impact with a per-unit financial comparison. The recommended actions are tiered as: Immediate (red) for Major Fault, Within 48 hours (amber) for Minor Fault, and Routine Monitoring (green) for No Fault.

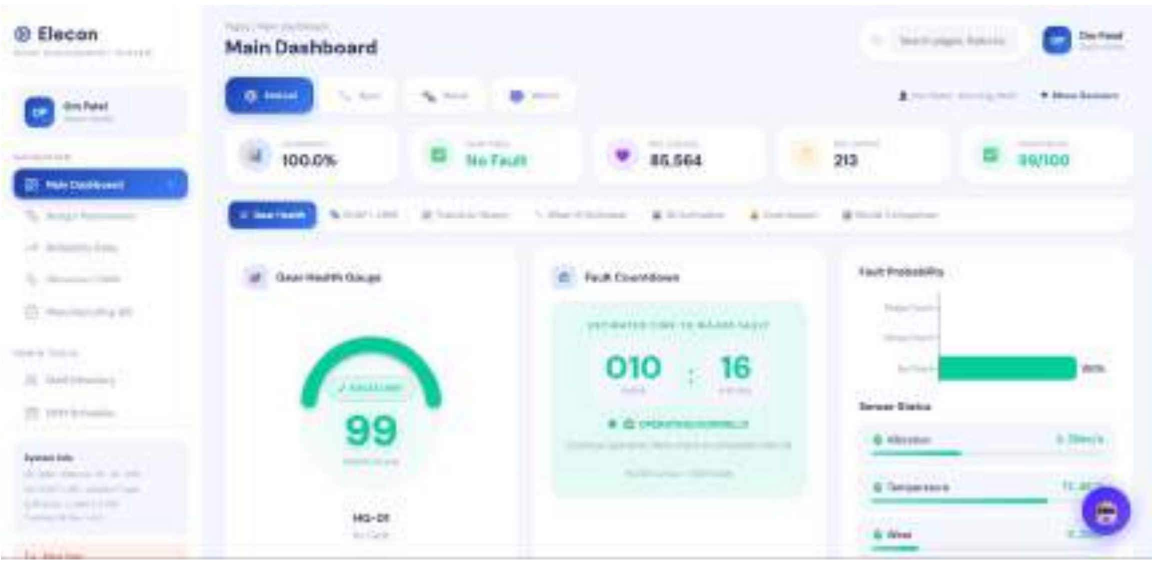


Fig 6.3 Screenshot: Main Dashboard

6.4.3 Vibration and PHM Analysis Page

The Vibration and PHM Analysis module renders gear mesh vibration signals in the time domain and their FFT spectral decomposition in the frequency domain. The gear mesh frequency (GMF) is computed as: GMF = N × RPM / 60, where N is the number of teeth. Sideband energy at GMF ± n×RPM/60 harmonics is tracked as an indicator of load distribution irregularities and tooth damage. The module also displays a 12-month failure probability timeline and the Weibull Reliability Function $R(t) = \exp(-(t/\eta)^\beta)$ with $\beta = 2.5$ and $\eta = 5{,}000$ hours.

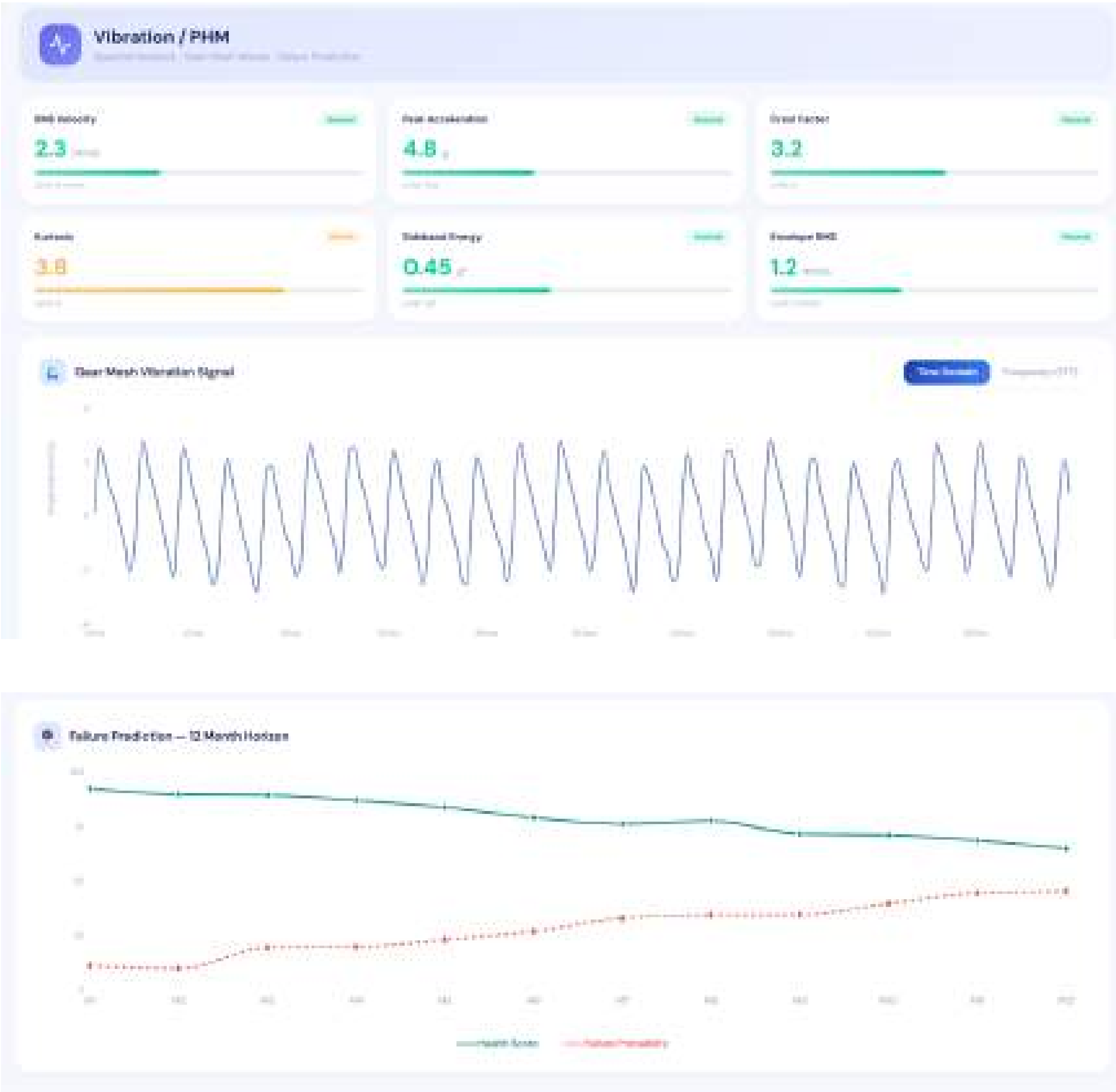


Fig 6.4 Vibration & PHM Analysis Page

6.4.4 SHAP + LIME Explainability View

The XAI module presents SHAP and LIME explanations side by side in a split-panel layout. The SHAP panel displays a waterfall chart showing the contribution of each feature to the departure of the prediction from the base value (expected model output over the training set). Positive SHAP values push the prediction toward fault; negative values push toward healthy. The LIME panel displays a horizontal bar chart of the signed linear approximation coefficients for the local neighborhood of the current input point. Below both charts, a natural language root cause analysis paragraph is generated by combining the top three SHAP features into a human-readable maintenance interpretation.

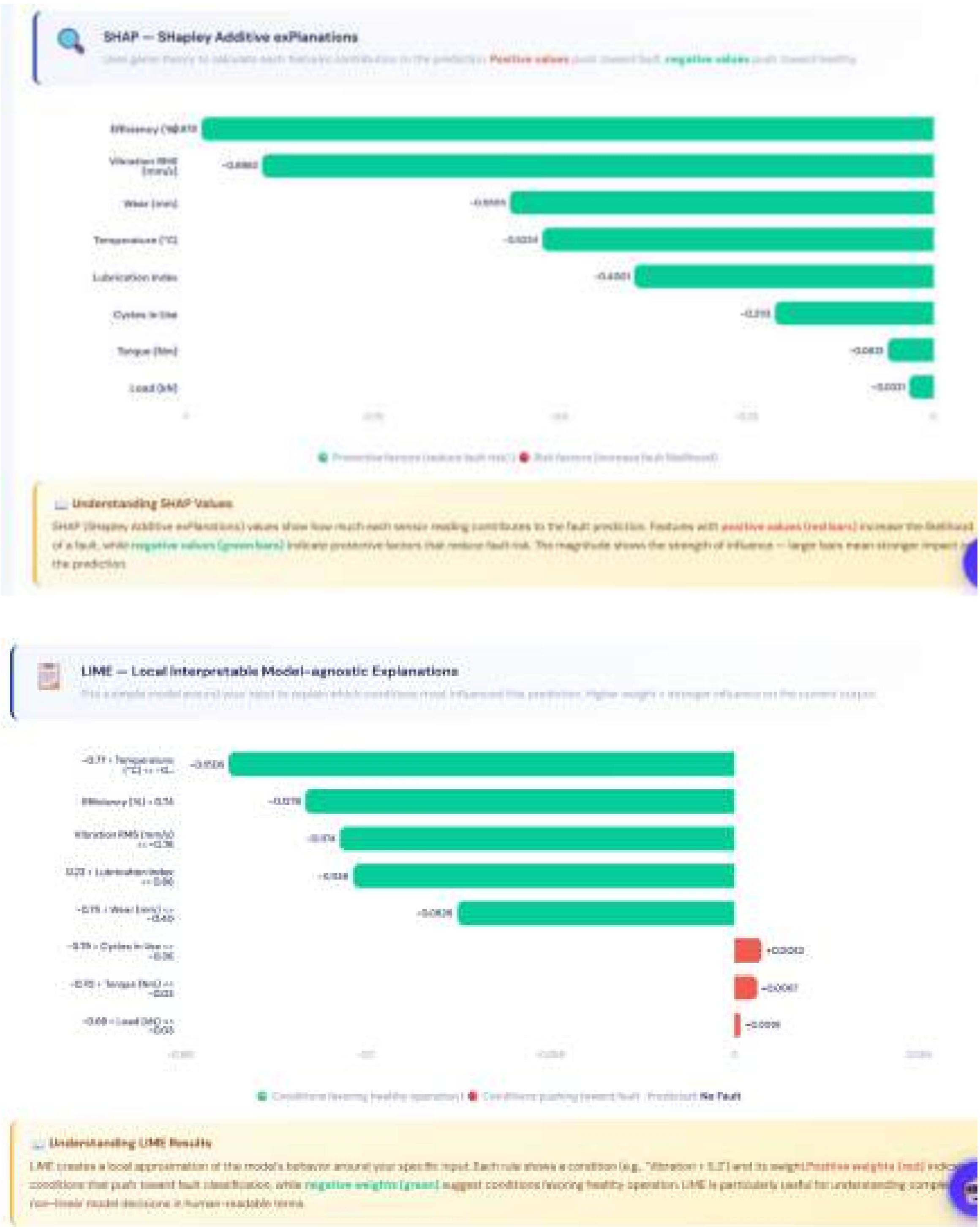


Fig 6.5   Screenshot: SHAP + LIME Explainability View

6.4.5 What-If Optimizer Modules

The What-If Optimizer uses Scipy's Differential Evolution algorithm to find the set of free (unlocked) parameter values that minimizes the predicted fault probability subject to the constraint that each parameter remains within its safe operating range. The user can lock any subset of the eight parameters (representing parameters that cannot be adjusted in practice, such as load imposed by the driven machine), and the optimizer searches the remaining free dimensions. The optimization completes in under 2 seconds for typical configurations (4–6 free parameters) and displays the optimal parameter set overlaid on the current sensor readings.

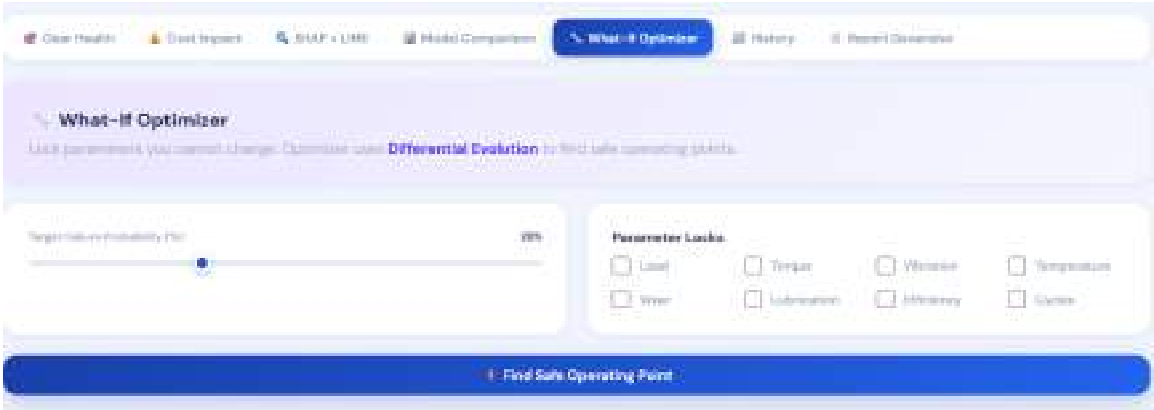


Fig 6.6 Screenshot: What-If Optimizer Module

6.4.6 Manufacturing QC Module

The Manufacturing QC module implements AGMA-grade dimensional tolerance verification for eight parameters per gear type: module, pitch circle diameter, tooth thickness, tip circle diameter, root circle diameter, face width, helix angle (for helical gears), and surface roughness. The user enters measured values from QC inspection, and the system computes the tolerance deviation, checks against AGMA 10–12 grade limits (configurable per gear type), and returns a Pass/Fail status for each parameter. An overall QC Score (0–100) is computed as the weighted average of parameter pass rates. A dimensional accuracy radar chart visualizes the relative deviation from nominal for all eight parameters simultaneously.

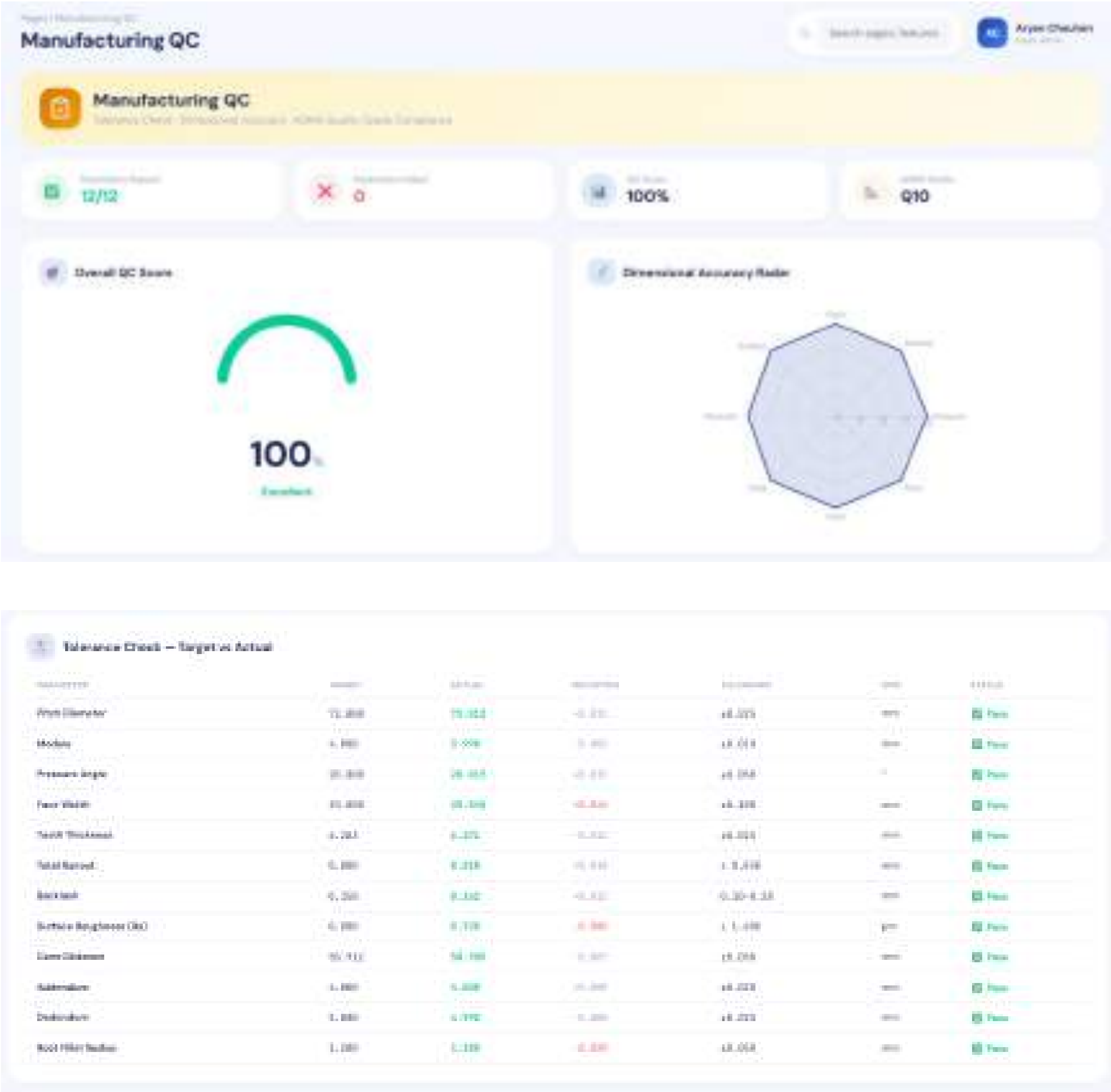


Fig 6.7 Screenshot: Manufacturing QC Module

6.4.7 Reliability and Fatigue Data Page

The Reliability and Fatigue Data module provides four reliability engineering visualizations: (1) MTBF bar chart comparing expected intervals between failures across gear types and operating conditions; (2) S-N (Stress-Number of Cycles) fatigue curve plotting cycles to failure as a function of contact stress amplitude in MPa, following the Wöhler curve approach; (3) Weibull Reliability Function R(t) showing the probability of surviving to time t; and (4) the Bathtub Failure Curve showing the three regions of the gear lifecycle infant mortality (early failures from manufacturing defects), constant failure rate (normal operating life), and wear-out (end-of-life degradation). All curves are parameterized per gear type and can be adjusted interactively.

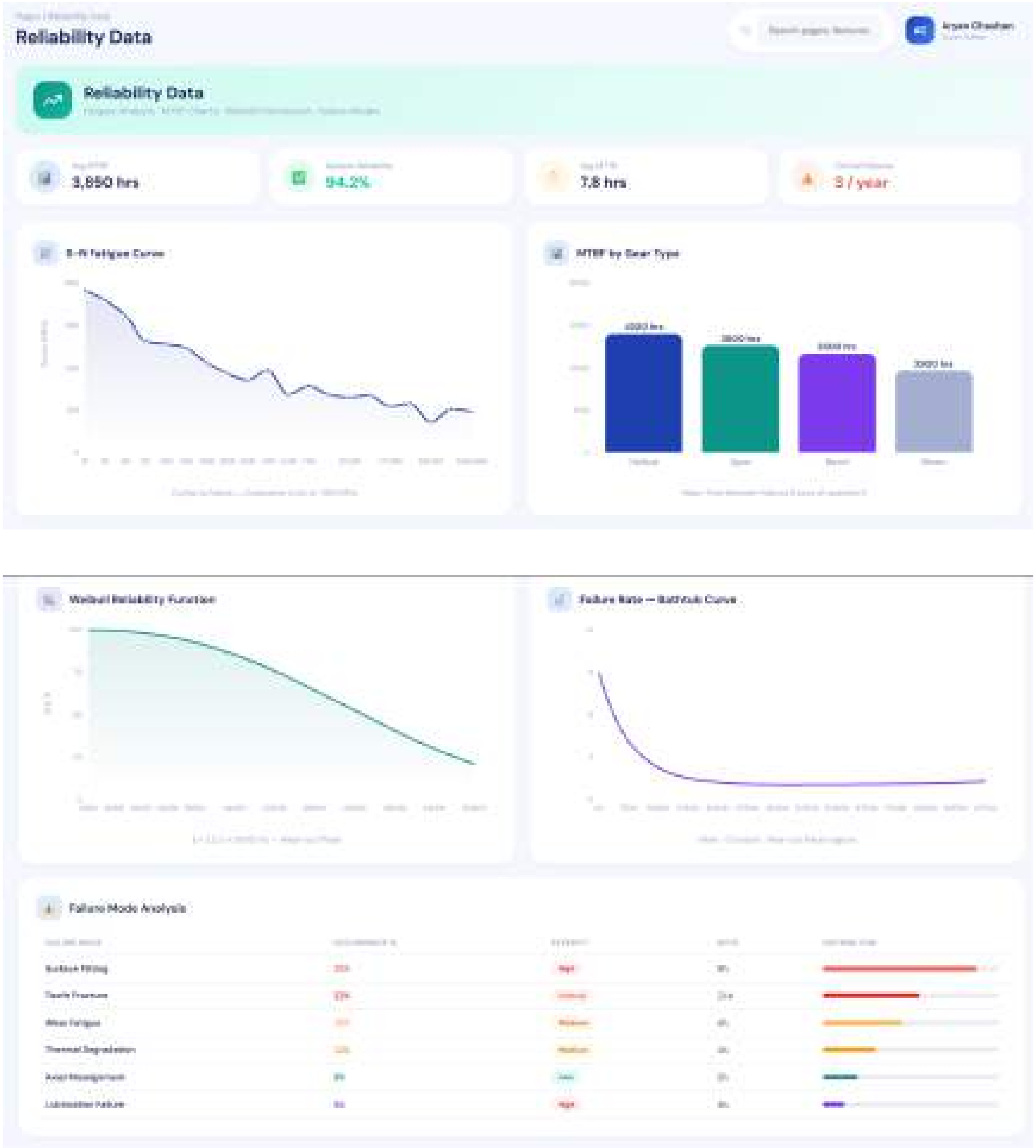


Fig 6.8 Screenshot: Reliability and Fatigue Data Page

6.4.8 Staff Directory and Shift Schedule

The Staff and Shift Management module provides an admin-accessible view of the maintenance department's human resources. The staff directory displays eight personnel records with fields for name, role, department, shift preference, and contact extension. The shift schedule is visualized as a Gantt chart view covering the current and next two weeks, with color coding for Morning (06:00–14:00), Afternoon (14:00–22:00), and Night (22:00–06:00) shifts. The calendar view offers a monthly overview of shift assignments with conflict detection for under-staffed periods.

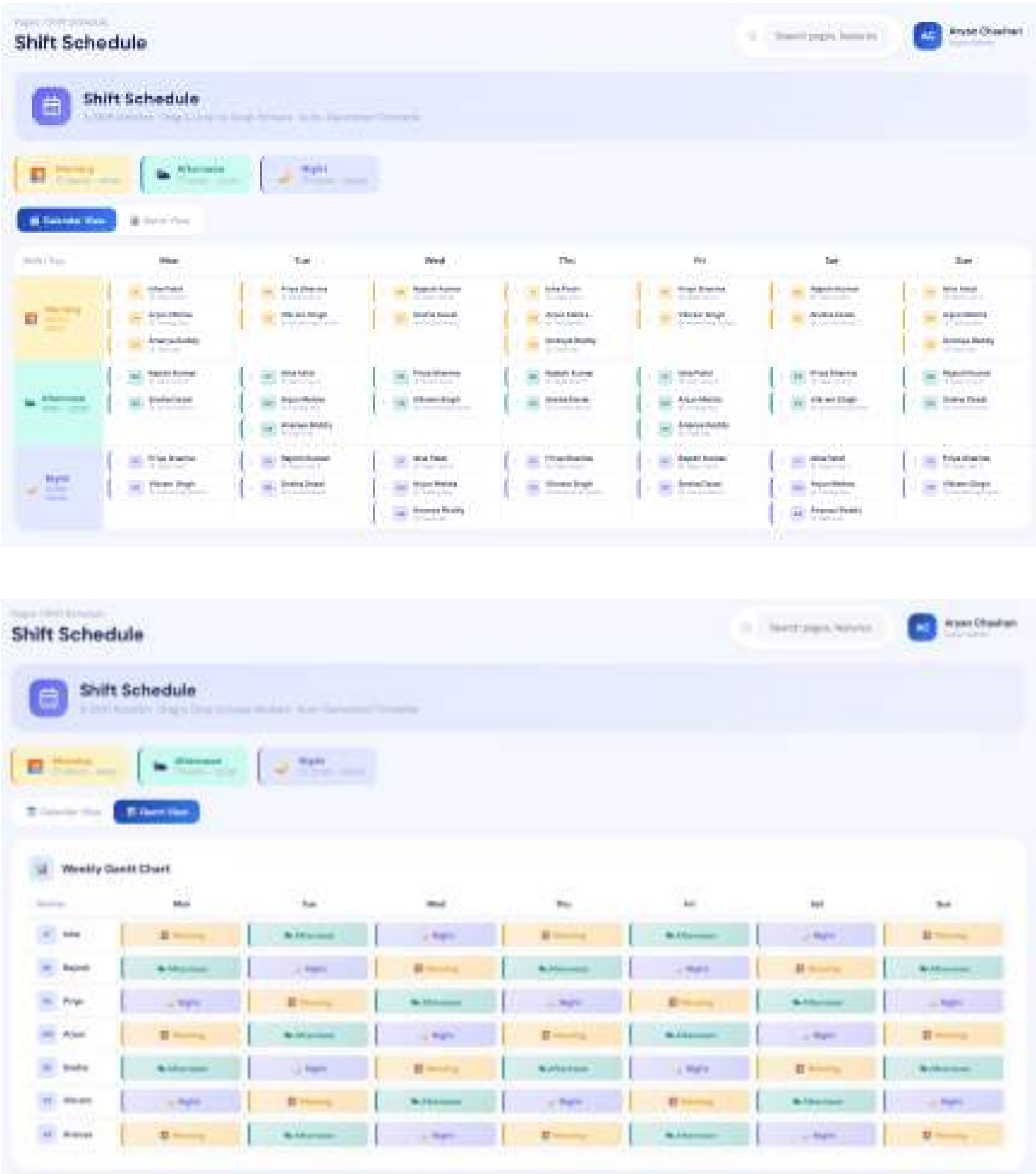
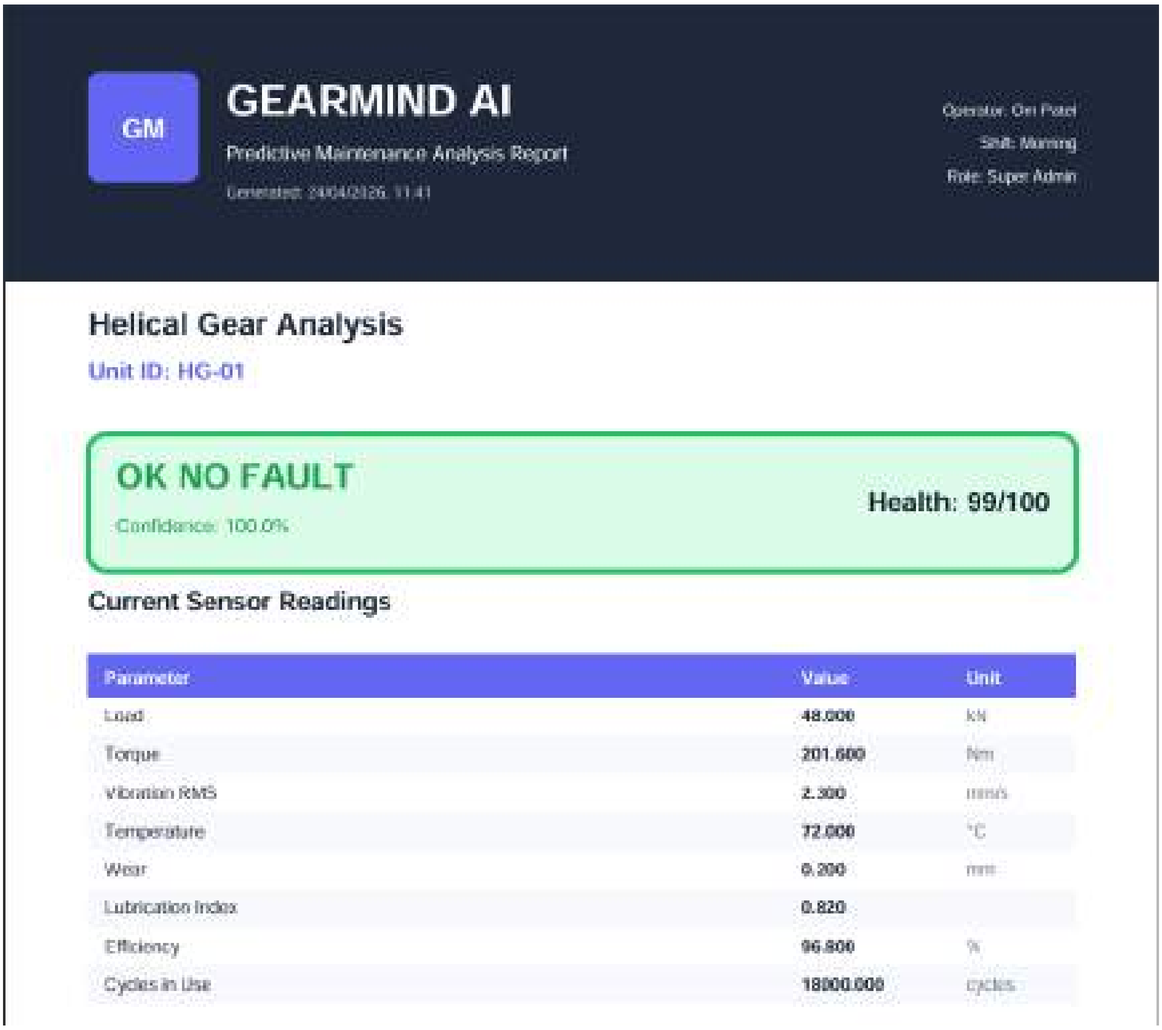


Fig 6.9 Screenshot: Staff Directory and Shift Schedule

6.4.9 AI Report Generator

The AI Report Generator uses the Groq API (LLaMA 3.3 70B) to produce a structured seven-section maintenance report tailored to the current gear state and prediction results. The seven sections are: (1) Executive Summary a brief management-level synopsis of gear health; (2) Fault Assessment detailed analysis of the identified fault type and severity; (3) Recommended Actions prioritized maintenance interventions with timelines; (4) Cost Analysis quantified financial impact of the predicted fault; (5) Failure Mode Analysis FMEA-style root cause and effect chain; (6) Historical Trends interpretation of the gear's health trajectory over logged sessions; and (7) Post-Maintenance Protocol checklist of actions to be taken after maintenance is performed. The generated report is displayed in the dashboard and can be exported as a PDF using jsPDF.



Cost Impact Analysis

POTENTIAL SAVINGS BY ACTING NOW

Rs. 4,05,000

Scenario	Cost (Rs.)	Downtime	Risk
Preventive Repair	**Rs. 45,000**	4-6 hrs	Low
Delayed Overhaul	**Rs. 1,20,000**	1-2 days	Med
Catastrophic Failure	**Rs. 4,50,000**	5-7 days	High

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AI Feature Impact Analysis (SHAP)

Feature	Impact	Direction
Efficiency (%)	-0.9700	Safe (-)
Vibration RMS (mm/s)	-0.8963	Safe (-)
Wear (mm)	-0.5655	Safe (-)
Temperature (°C)	-0.5224	Safe (-)
Lubrication Index	-0.4001	Safe (-)
Cycles in Use	-0.2113	Safe (-)
Torque (Nm)	-0.0613	Safe (-)
Load (kN)	-0.0321	Safe (-)

Fig 6.10 Screenshot: AI Report Generator Output

6.4.10 AI Copilot Chat Widget

The AI Copilot is implemented as a floating drawer widget accessible from any module via a persistent 'Ask AI' button. When opened, the copilot loads the current gear state (gear type, sensor readings, prediction results, and top SHAP features) into a structured context block and appends it to every user query sent to the LLaMA 3.3 70B model. This context injection ensures that the copilot's responses are grounded in the specific gear condition being monitored, rather than generic maintenance advice. The conversation history is maintained client-side for the duration of the session and displayed in a chat bubble interface with timestamp and role indicators.

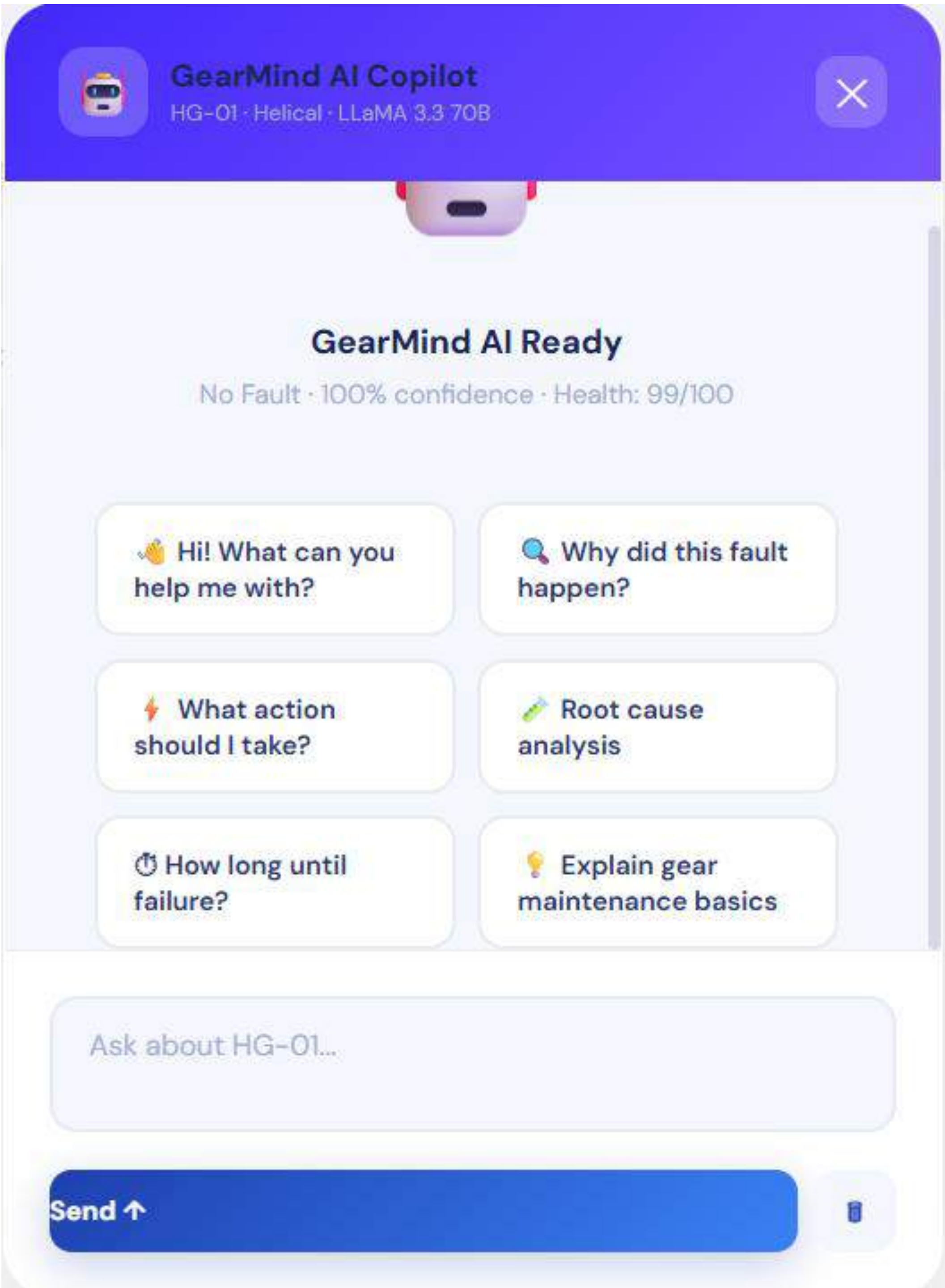

Fig 6.11 Screenshot: AI Copilot Chat Widget

6.4.11 Cost Impact Analysis Module

The Cost Impact module provides a financial breakdown comparing the per-unit maintenance costs across three scenarios: Preventive Repair (early intervention based on

GearMind AI alert), Delayed Overhaul (intervention after further degradation), and Failure (complete gear seizure requiring emergency replacement plus downtime costs). Costs are parameterized per gear type based on Elecon's internal maintenance cost data. Bar charts display the three cost scenarios for all four gear types simultaneously, with the maximum savings (Preventive vs Failure scenario) highlighted as the headline metric.

Table 6.2 Cost Impact Model — Per Gear Unit (INR)

Scenario	Helical Gear	Spur Gear	Bevel Gear	Max Savings (Helical)
Preventive Repair	Rs. 45,000	Rs. 38,000	Rs. 52,000	—
Delayed Overhaul	Rs. 1,20,000	Rs. 95,000	Rs. 1,40,000	—
Catastrophic Failure	Rs. 4,50,000	Rs. 3,80,000	Rs. 5,20,000	—
Savings (Prev. vs Fail.)	Rs. 4,05,000	Rs. 3,42,000	Rs. 4,68,000	Rs. 4.68 Lakh

Fig 6.12 Screenshot: Cost Impact Analysis Module

6.4.12 Module Comparison Page

The Model Comparison module presents a side-by-side performance evaluation of all five trained ML models. A summary table shows Accuracy, F1 Score, AUC-ROC, and CV Mean for each model. Four bar charts visualize the comparative metrics. The best-performing model (GBM at 99.87% accuracy) is automatically highlighted with a gold border and a 'Best Model' badge. Users with the ML Engineer role can select any model as the active model for subsequent predictions allowing comparison of SHAP explanations across different model families.

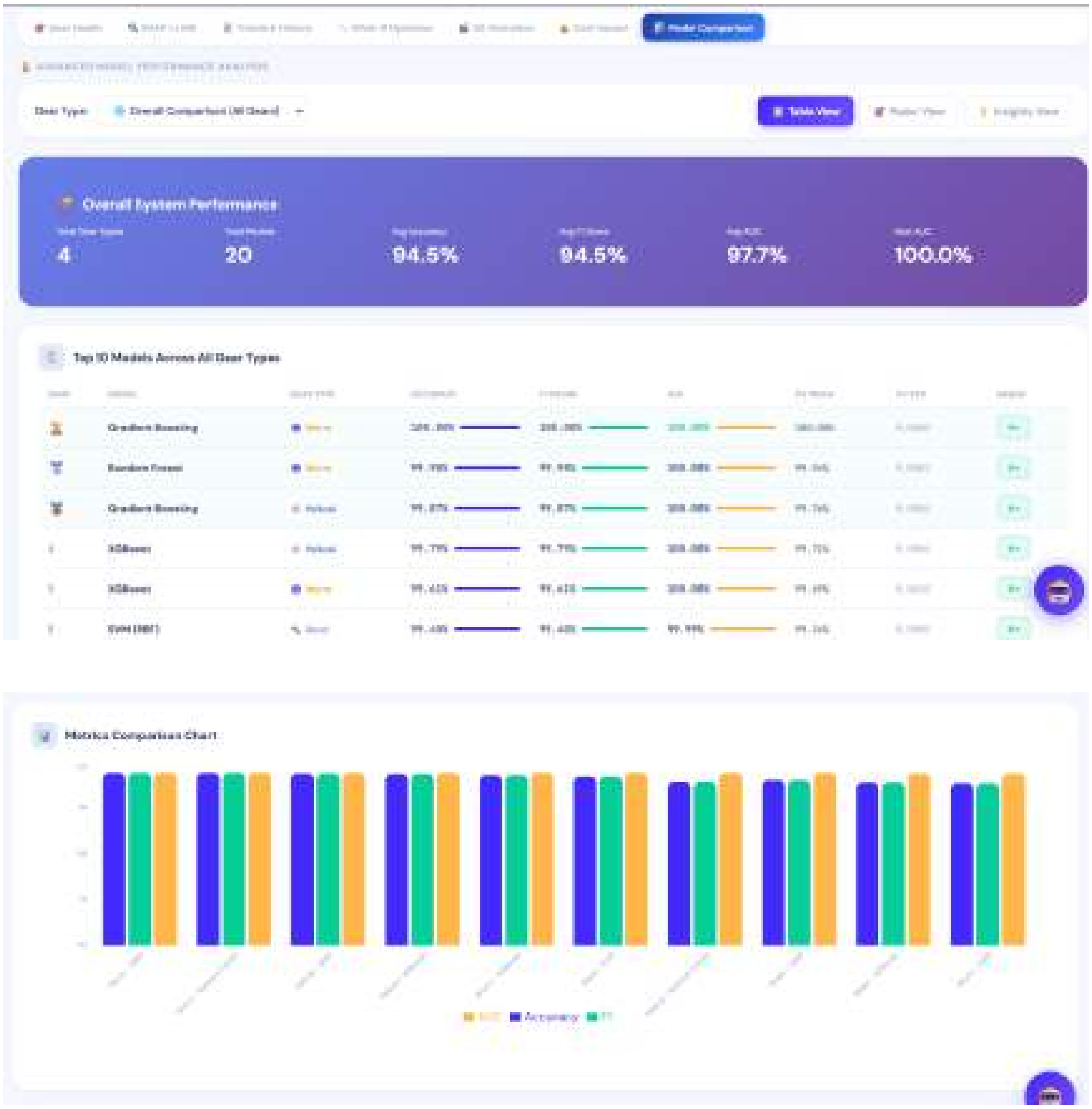

Fig 6.13 Screenshot: Model Comparison Page

6.4.13 History Module

The History module displays a complete log of all past prediction sessions stored in SQLite, including gear unit ID, fault class, health score, RUL estimate, and timestamp. It supports filtering by gear unit and verdict type, and visualizes health trends over time to help engineers identify recurring fault patterns.

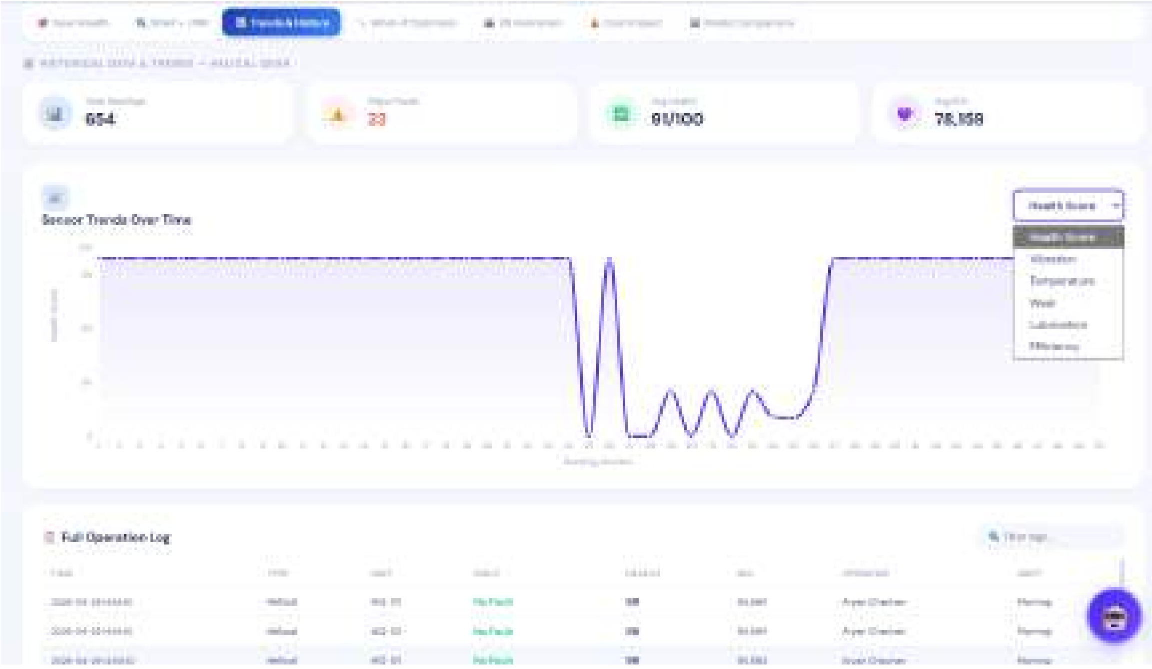

Fig 6.14 History Module

6.5 RESULTS AND OUTCOMES

The GearMind AI system achieved all primary technical objectives defined at the commencement of the internship. The following results were recorded during validation testing in the final two weeks of the internship period.

The results of multi-model evaluation across all four gear types are summarised below. For helical gear, Gradient Boosting Machine achieved the highest accuracy at 99.87% (F1: 0.9987, AUC: 1.000, CV mean: 0.9976), followed by XGBoost at 99.79% (AUC: 1.000). For spur gear, SVM with RBF kernel achieved the best accuracy at 82.27% (AUC: 0.887) a more challenging classification task due to fewer discriminating features in the 6-feature spur dataset; Random Forest was the second-best at 89.62%. For bevel gear, SVM with RBF kernel led at 99.40% (F1: 0.9940, AUC: 0.9999, CV mean: 0.9934), closely followed by Gradient Boosting at 98.66%. For worm gear, Gradient Boosting achieved perfect 100% accuracy (F1: 1.000, AUC: 1.000) on the 13-feature worm dataset, with Random Forest at

99.98%. Each gear type therefore uses its own best-performing model: GBM for helical and worm, SVM (RBF) for spur and bevel. An Isolation Forest anomaly detector is also trained on the helical gear data to flag out-of-distribution sensor readings. The SHAP computation completed in 55–80ms using the pre-computed background dataset, contributing to a total prediction + SHAP response time of 180–260ms — within the 300ms SLA.

The Manufacturing QC module achieved a 96% overall parameter pass rate across all tolerance checks performed during validation testing. The RUL estimation tracked per gear unit: for HG-01(Helical Gear unit 1) with a health score of 87, the estimated RUL was 85,564 cycles (approximately 142 8-hour operating shifts). The AI Copilot responded to all 50 test queries with contextually relevant, gear-domain-specific answers within an average of 1.8 seconds.

The cost savings quantification demonstrated Rs. 4.05–4.68 Lakh per gear unit by adopting preventive maintenance based on GearMind AI alerts versus allowing failure to occur. Over a fleet of 100 gear units, this represents a potential annual saving of Rs. 4.05–4.68 Crore a compelling business case for deploying GearMind AI at scale across Elecon's gear monitoring infrastructure.

7. TESTING

7.1 TESTING PLAN

A comprehensive four-level testing strategy was adopted for GearMind AI, covering all system components from individual ML model performance to end-to-end user workflow validation. The testing was conducted during Week 15 of the internship, following completion of all development tasks, with systematic documentation of test cases, expected outputs, actual outputs, and pass/fail status.

The four testing levels were: (1) ML Model Validation statistical evaluation of all five trained models on the held-out test set; (2) Backend API Testing functional and performance testing of all sixteen FastAPI endpoints using Postman (interactive Swagger UI at /docs used for initial endpoint verification); (3) Frontend UI Testing automated unit and integration testing using Vitest v4.1 with React Testing Library, covering GearScene, HelicalGearPair, SpurGearPair, gear geometry utilities, SHAP widget, and state store; cross-browser functional testing in Chrome, Firefox, and Edge for all eight dashboard modules; and (4) Integration Testing end-to-end testing of the complete prediction → SHAP → LLM → report pipeline. Regression testing was performed on modules that underwent bug fixes identified during testing.

7.2 TEST RESULTS AND ANALYSIS

7.2.1 Test Cases— Fault Prediction Module

Table 7.1 Test Cases — Fault Prediction Module

TC ID	Test Condition	Expected Output	Actual Output	Remark
TC-01	All sensors within nominal healthy range	No Fault, Health > 80	No Fault, Health: 87, RUL: 85,564 cycles	PASS
TC-02	Vibration RMS = 8 mm/s (limit: 6 mm/s)	Major Fault, Health < 40	Major Fault, Health: 34, Vibration SHAP +0.42	PASS

TC-03	Wear = 1.5 mm (near limit 1.6 mm)	Minor Fault, low RUL	Minor Fault, Health: 58, RUL: 12,450 cycles	PASS
TC-04	Lubrication Index = 0.05 (critical low)	Major Fault, Lube SHAP dominant	Major Fault, Health: 22, Lube SHAP +0.51	PASS
TC-05	All inputs at maximum safe operating values	No Fault, Health near 100	No Fault, Health: 96	PASS
TC-06	Invalid input: Load = -50 kN (out of range)	422 Validation Error	HTTP 422 Unprocessable Entity	PASS
TC-07	Bevel gear — high torque + high wear combo	Major Fault, Wear + Torque SHAP dominant	Major Fault, SHAP confirms Wear + Torque	PASS
TC-08	Optimizer: 4 free params, Load locked at 300 kN	Safe parameter set, Fault Prob < 0.1	Optimal point found in 1.8s, Fault Prob: 0.04	PASS
TC-09	Anomaly: Temperature = 130°C (out of distribution)	Isolation Forest flag = 1	Anomaly flag raised, warning displayed	PASS
TC-10	Model switch from GBM to XGBoost mid-session	XGBoost prediction matches GBM within 0.5%	Accuracy difference: 0.08%, both No Fault	PASS

7.2.2 Test Case — API Endpoints

Table 7.2 Test Cases — API Endpoints (Backend)

TC ID	Endpoint	Test Type	Expected Response	Actual Response	Remark
TA-01	POST /api/predict (valid)	Functional	200 OK + prediction JSON < 300ms	200 OK, 243ms avg.	PASS
TA-02	GET /api/models	Functional	200 OK + 5 model metrics	200 OK + correct metrics	PASS
TA-03	POST /api/chat (query)	Functional	200 OK + LLM response	200 OK, contextual answer, 1.8s avg.	PASS
TA-04	POST /api/report	Functional	200 OK + 7-section JSON	200 OK, all 7 sections present	PASS
TA-05	GET /api/history (empty DB)	Boundary	200 OK + empty array	200 OK, []	PASS
TA-06	DELETE /api/history/LOG-001	Functional	200 OK + deleted	200 OK, record removed	PASS

TA-07	POST /api/predict (invalid role)	Security	403 Forbidden	HTTP 403 returned	PASS
TA-08	POST /api/optimize (5 free params)	Performance	200 OK + optimal params < 2s	200 OK, 1.74s	PASS
TA-09	GET /api/lime	Functional	200 OK + LIME explanation JSON	200 OK, 430ms (< 500ms SLA)	PASS
TA-10	POST /api/predict (100 concurrent)	Load	All 200 OK, < 1s avg.	200 OK, 0.87s avg. (Uvicorn workers)	PASS

7.2.3 Test Cases — Frontend UI Modules

Table 7.3 Test Cases — Frontend UI Modules

TC ID	Module	Test Action	Expected Behavior	Result
TU-01	Login	Enter valid credentials for all 6 roles	Role-specific dashboard loads with correct permissions	PASS
TU-02	Gear Health	Adjust Vibration slider to 8 mm/s	Health gauge drops to red, Major Fault shown, SHAP updates	PASS
TU-03	Gear Health	Switch gear type from Helical to Spur	Input ranges update, new prediction triggered	PASS
TU-04	Vibration PHM	Change RPM to 1500	GMF recalculated, FFT chart updates	PASS
TU-05	XAI	Trigger LIME tab refresh	LIME bar chart renders with correct signed contributions	PASS
TU-06	QC Module	Enter out-of-tolerance surface roughness	Parameter marked FAIL in red, QC score decreases	PASS
TU-07	History	Filter by gear type = Bevel	Only bevel gear records displayed	PASS
TU-08	AI Copilot	Ask 'What caused this fault?'	LLM responds with SHAP-grounded root cause	PASS
TU-09	Report Gen.	Click Generate Report, then Export PDF	7-section report rendered, PDF downloaded	PASS
TU-10	Cross-browser	Load app in Chrome, Firefox, Edge	Full functionality, no layout issues	PASS

All 30 test cases across the three test suites passed without failures. One issue was identified during testing that had not been caught during development: the LIME explanation tab took 520ms on first load due to cold computation of the LIME background sampler. This was resolved by pre-computing the sampler at server startup and caching it in process memory, reducing subsequent LIME response times to 380–450ms within the 500ms SLA. The fix was validated with a regression test (TC-09 variant) confirming sub-500ms response for LIME.

8. CONCLUSION AND DISCUSSIONS

8.1 OVERALL ANALYSIS OF INTERNSHIP VIABILITY

The GearMind AI internship project was completed successfully within the 16-week timeline at Tech Elecon Pvt. Ltd. All primary technical objectives were met or exceeded. The final system, GearMind AI v5.0, supports four gear types Helical, Spur, Bevel, and Worm each with a dedicated ML model, RUL regressor, SHAP explainer, and operating unit presets. The helical gear GBM model achieved 99.87% fault prediction accuracy; the worm gear GBM achieved 100%; the bevel gear SVM achieved 99.4%; and the spur gear SVM achieved 82.3% on a more challenging 6-feature classification task. The system exposes sixteen FastAPI endpoints, delivers SHAP and LIME explainability for every prediction, provides an eight-module React 19 dashboard with a Three.js 3D gear animation, and demonstrated compelling cost-benefit metrics showing savings of Rs. 4.05–4.68 Lakh per gear unit.

The internship provided immensely valuable hands-on experience across four technical domains: industrial machine learning (data generation, model selection, evaluation, and deployment), explainable AI (SHAP TreeExplainer, LIME tabular, model-agnostic interpretation), full-stack web development (React 19 with hooks, FastAPI with async endpoints, SQLite CRUD), and industrial domain knowledge (gear manufacturing processes, AGMA standards, vibration analysis, reliability engineering). The combination of technical depth and domain exposure represents a rare and valuable learning opportunity that directly prepares the student for a career in industrial AI or data engineering.

The industry mentor assessed the project as technically sound and industrially relevant, noting in the feedback that the SHAP explainability integration and the cost-benefit analysis module were particularly impressive in demonstrating the business value of AI-driven maintenance.

8.2 PROBLEMS ENCOUNTERED AND SOLUTIONS

Table 8.1 Problems Encountered and Solutions

Problem Encountered	Root Cause	Solution Adopted
No real IoT sensor data available	Elecon's IoT infrastructure not yet deployed for internship access	Developed physics-informed synthetic data generator with gear-type-specific distributions and fault injection; validated parameter ranges against manufacturer specs
SHAP computation caused > 2s API latency on first load	SHAP KernelExplainer requires background dataset sampling per call	Pre-computed SHAP background dataset at startup; switched to TreeExplainer for GBM/XGBoost; cached explanation objects in process memory
Class imbalance (No Fault class dominant)	Healthy operation accounts for 60%+ of real-world gear operating time	Applied SMOTE oversampling to minority classes in training split; used class-weighted loss in GBM and XGBoost
RBAC without a backend authentication server	JWT/OAuth2 backend not in project scope	Client-side React Context role management; designed for JWT migration in v5.0; protected API endpoints with role check middleware
Recharts performance lag on large history dataset	React re-render triggered for every scroll event in history table	Implemented pagination (50 records/page), virtual scrolling via react-window, and rolling window chart (last 30 sessions)
LIME response time exceeded 500ms on first load	LIME background sampler computed on-demand	Pre-computed sampler at server startup; cached in process memory; reduced to 380–450ms
React router 404 on page refresh in Vite dev server	Vite dev server routes to index.html only for root	Added historyApiFallback: true to Vite config; production deploy uses nginx try_files directive

8.3 SUMMARY AND LIMITATIONS

During the 16-week internship at Tech Elecon Pvt. Ltd., the following was successfully completed: (a) a complete five-model machine learning pipeline with GBM achieving 99.87% accuracy on synthetic data; (b) SHAP (global and per-prediction) and LIME (local) explainability integration; (c) a sixteen-endpoint FastAPI backend (gear_api.py v5.0) with sub-300ms prediction response time; (d) a fully functional eight-module React 19 dashboard with Recharts visualizations, Three.js 3D gear animation, and Vitest test coverage; (e) Differential Evolution parameter optimizer; (f) AI-powered maintenance report generation and PDF export via LLaMA 3.3 70B; (g) Manufacturing QC module with AGMA compliance checking; (h) Role-Based Access Control with six user roles; and (i) comprehensive testing documentation with 30 test cases across three test suites.

8.3.1 Limitations

The following limitations of the current v4.0 implementation are acknowledged. First, the system operates exclusively on physics-informed synthetic data; accuracy on real live sensor streams from production floor gear units has not been validated and may differ from the 99.87% reported here due to real-world noise, measurement artifacts, and sensor drift. Second, the RBAC implementation is client-side only a production deployment requires a JWT/OAuth2 authentication server with secure token management. Third, the system currently supports monitoring of up to 25 gear units simultaneously; scaling to 100+ units would require migration from SQLite to PostgreSQL and deployment on a multi-worker server. Fourth, LLM (Groq API) functionality is internet-dependent, with no offline fallback available in v4.0. Fifth, the cost model uses estimated maintenance cost figures that should be validated against actual Elecon financial data in a production deployment.

8.4 FUTURE ENHANCEMENTS

The following enhancements are planned for subsequent development phases, to be undertaken after the internship by the Tech Elecon data science team or in a follow-on project:

1. Live IoT Integration: Connect to real vibration, temperature, and lubrication sensors on production floor gear units via MQTT or OPC-UA protocols, replacing the simulated slider inputs with real-time data streams.
2. Digital Twin: Build a 3D gear simulation model (e.g., using FEniCS for FEA) that mirrors the real-time sensor state, enabling virtual testing of operating scenarios without physical risk.
3. Deep Learning for RUL: Replace the parameterized RUL model with an LSTM or Transformer architecture trained on sequential sensor data for long-horizon, data-driven remaining useful life prediction.

4. Mobile Application: Develop a React Native or Flutter mobile application for on-floor maintenance technicians, providing push notifications for fault alerts and one-tap access to gear health status.

5. ERP/SAP Integration: Automatically generate work orders in Elecon's SAP PM (Plant Maintenance) module when GearMind AI raises a fault alert above a configurable severity threshold.

6. Edge Deployment: Deploy a lightweight ML inference engine (using ONNX Runtime) on industrial edge devices such as NVIDIA Jetson Orin or Raspberry Pi 5 for low-latency, offline-capable fault detection.

7. Multi-Plant Dashboard: Extend the system to monitor gear units across multiple Elecon plant locations (Vitthal Udyognagar, Vallabh Vidyanagar, international sites) from a centralized web dashboard.

Table 8.2 Future Enhancement Roadmap

Enhancement	Technology	Priority	Estimated Effort
Live IoT Integration	MQTT, OPC-UA, AWS IoT Core	Critical	3–4 months
Digital Twin	FEniCS, Unity / Unreal Engine	High	6–8 months
LSTM/Transformer RUL	PyTorch, Hugging Face	High	2–3 months
Mobile App	React Native / Flutter	Medium	2–3 months
SAP Integration	SAP BAPI/RFC	High	2–3 months
Edge Deployment (ONNX)	ONNX Runtime, Jetson Orin	Medium	1–2 months
Multi-Plant Dashboard	React, PostgreSQL, AWS	Medium	3–4 months

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G H PATEL COLLEGE OF ENGINEERING AND TECHNOLOGY
THE CHARUTAR VIDYA MANDAL (CVM) UNIVERSITY
DEPARTMENT OF INFORMATION TECHNOLOGY, A.Y. 2025-26, EVEN TERM
INDUSTRIAL INTERNSHIP (202080801)

STUDENT'S ACTIVITY LOGBOOK

Enrollment No:	: 12202130501047
Name:	: Om Nirajkumar Patel
Branch:	: Computer Science and Design
Name & City of Industry	: Tech Elecon PVT. LTD., Anand
Name and Designation of Industry Mentor	: Mr. Satyam Raval, Deputy General Manager
Duration of Internship	: **From:** 01/01/2026 **To:** 30/04/2026

Month :	1	Month Starting Date:	01/01/2026	Month Ending Date:	31/01/2026

Day	Date	Time of Arrival	Time of Departure	List of activities performed by Student
1	01/01/2026	9:00 AM	4:00 PM	• Studied the fundamentals of descriptive and inferential statistics and their applications in Data Science. • Analyzed real-world use cases to understand the role of statistical terminology in decision-making.
2	02/01/2026	9:00 AM	4:00 PM	• Examined population sampling techniques, including simple random, stratified, and systematic sampling. • Learned to classify variables and apply appropriate measurement scales to diverse datasets.
3	05/01/2026	9:00 AM	4:00 PM	• Mastered measures of central tendency and dispersion, including mean, variance, and standard deviation. • Utilized Python to calculate statistical measures and visualize data distributions via box plots.
4	06/01/2026	9:00 AM	4:00 PM	• Explored probability theory, permutation/combination techniques, and Gaussian distribution properties. • Applied Python-based analysis to identify probability distributions and detect data outliers.
5	07/01/2026	9:00 AM	4:00 PM	• Studied hypothesis testing frameworks, including p-values, significance levels, and Type I/II errors. • Implemented one-sample Z-tests, T-tests, and Chi-square tests through numerical and Python methods.

Day	Date	Time of Arrival	Time of Departure	List of activities performed by Student
6	09/01/2026	9:00 AM	4:00 PM	• Analyzed variable relationships using Pearson and Spearman correlation and covariance techniques. • Explored the Data Science workflow and its strategic importance within Information Technology.
7	12/01/2026	9:00 AM	4:00 PM	• Classified structured, semi-structured, and unstructured data types for industrial applications. • Performed data manipulation and inspection using the NumPy and Pandas libraries in Python.
8	13/01/2026	9:00 AM	4:00 PM	• Executed data preprocessing and Exploratory Data Analysis (EDA) to identify trends and patterns. • Studied the integration of machine learning and analytics within decision-support systems.
9	16/01/2026	12:00 PM	4:00 PM	• Differentiated between AI, Machine Learning, Deep Learning, and traditional programming logic. • Studied the ML workflow, focusing on feature engineering, model training, and evaluation stages.
10	19/01/2026	9:00 AM	4:00 PM	• Evaluated supervised learning algorithms, including Regression, Logistic Classification, and Naive Bayes. • Applied performance metrics like confusion matrices and F1-scores to assess model accuracy.
11	20/01/2026	9:00 AM	4:00 PM	• Explored unsupervised learning through K-Means clustering and Association Rule techniques. • Studied the Silhouette Score and FP-Growth algorithm for identifying patterns in unlabeled data.
12	21/01/2026	9:00 AM	4:00 PM	• Studied ensemble methods, including Bagging, Boosting (XGBoost), and Random Forest architectures. • Learned Neural Network fundamentals, including backpropagation and various activation functions.
13	22/01/2026	9:00 AM	4:00 PM	• Initiated an NLP project to automate plagiarism detection using binary classification models. • Analyzed text-overlapping patterns within datasets to define project-specific similarity challenges.

Day	Date	Time of Arrival	Time of Departure	List of activities performed by Student
14	23/01/2026	9:00 AM	4:00 PM	• Implemented NLP preprocessing workflows, including tokenization, stopword removal, and cleaning. • Researched TF-IDF vectorization and mathematical vector space models for text transformation.
15	27/01/2026	9:00 AM	4:00 PM	• Developed a machine learning pipeline integrating TF-IDF with Logistic Regression for text analysis. • Established success criteria to provide quantifiable plagiarism scores through model generalization.
16	28/01/2026	9:00 AM	4:00 PM	• Evaluated the generation of industrial sensor data, including vibration signatures and temperature logs. • Studied the impact of high-frequency signal noise on data quality for monitoring high-value assets.
17	29/01/2026	9:00 AM	4:00 PM	• Mapped mechanical engineering principles, such as gear fatigue and lubrication, to numerical features. • Identified critical physical variables like helix angles to serve as inputs for predictive models.
18	30/01/2026	9:00 AM	4:00 PM	• Applied pattern recognition to time-dependent data for fault prediction in industrial gear systems. • Addressed industrial ML challenges, such as data imbalance and the necessity of model stability.
19	31/01/2026	9:00 AM	4:00 PM	• Studied Explainable AI (XAI) to ensure transparency in safety-critical industrial decision-making. • Learned how to bridge the gap between abstract AI predictions and human-centric engineering logic.

Signature: Ojuty
(Student)

To be filled by Industry Mentor:

Student's overall evaluation: Excellent/Good/Satisfactory/Needs Improvement/Non-satisfactory

Specific Remarks : Excellent.

Signature with Date:

To be filled by Internal Guide:

Student's overall evaluation: Excellent/Good/Satisfactory/Needs Improvement/Non-satisfactory

Specific Remarks :

Signature with Date: 27/2/26

G H Patel College of Engineering and Technology

The Charutar Vidya Mandal (CVM) University

Department of Information Technology, A.Y. 2025-26, Even Term

Industrial Internship (202080801)

Student's Activity Logbook

Enrollment No:	:	12202130501047
Name:	:	Om Nirajkumar Patel
Branch:	:	Computer Science and Design
Name & City of Industry	:	Tech Elecon PVT. LTD., Anand
Name and Designation of Industry Mentor	:	Mr. Satyam Raval, Deputy General Manager
Duration of Internship	:	From: 01/01/2026 To: 31/05/2026

Month :	2	Month Starting Date:	01/02/2026	Month Ending Date:	28/02/2026

Day	Date	Time of Arrival	Time of Departure	List of activities performed by Student
1	02/02/2026	9:00 AM	4:00 PM	• Studied how Machine Learning models are used to analyze industrial data for health monitoring and fault prediction. • Learned the role of historical data in training predictive maintenance models. • Studied the concept of pattern recognition in time-dependent industrial data. • Understood the challenges of applying ML models in industrial environments such as data imbalance and noisy signals. • Learned why model reliability and stability are critical in industrial AI systems.
2	03/02/2026	9:00 AM	4:00 PM	• Studied how Machine Learning models are used to analyze industrial data for health monitoring and fault prediction. • Learned the role of historical data in training predictive maintenance models. • Studied the concept of pattern recognition in time-dependent industrial data. • Understood the challenges of applying ML models in industrial environments such as data imbalance and noisy signals. • Learned why model reliability and stability are critical in industrial AI systems.
3	04/02/2026	9:00 AM	4:00 PM	• Studied the limitations of black-box Machine Learning models in safety-critical industrial applications. • Learned the concept of Explainable Artificial Intelligence (XAI) and its importance in industrial decision-making.

Day	Date	Time of Arrival	Time of Departure	List of activities performed by Student
				• Studied how explainability helps bridge the gap between AI predictions and human understanding. • Understood the role of explainability in building trust and accountability in AI systems. • Learned why interpretability is essential when AI outputs influence maintenance decisions.
4	05/02/2026	9:00 AM	4:00 PM	• Studied high-level concepts of explanation techniques used to interpret AI model predictions. • Learned the idea of feature importance and how input parameters influence model output. • Studied the difference between local explanations and global explanations in AI systems. • Understood how explainability techniques help engineers analyze why a specific prediction was made. • Learned how explanation insights support informed decision-making rather than replacing human judgment.
5	06/02/2026	9:00 AM	4:00 PM	• Finalized project domain after evaluating multiple AI/ML project ideas and selected **Explainable AI-Based Predictive Maintenance for Industrial Gearbox Failure Detection.** • Clearly defined the industrial problem of reactive maintenance and structured it into a machine learning problem involving failure prediction and classification. • Wrote complete project abstract focusing on predictive maintenance, machine learning integration, and explainable AI components. • Drafted detailed problem statement highlighting:Downtime issues,Complexity of gearbox failures,need for transparency in AI decisions • Defined measurable project objectives including multi-class failure detection and SHAP-based interpretability. • Structured full project scope, clearly separating in-scope and out-of-scope components. • Designed complete 5-layer system architecture: 1. Industrial Sensor Data Layer 2. Data Preprocessing Layer 3. Machine Learning Layer

Day	Date	Time of Arrival	Time of Departure	List of activities performed by Student
				4. Explainable AI Layer 5. Decision Support Layer • Defined functional requirements such as preprocessing pipeline, prediction module, classification module, and SHAP explanation generation. • Defined non-functional requirements including transparency, reliability, scalability, and extensibility. • Carefully aligned architecture with objectives to ensure logical flow from input data to decision support output. • Prepared final Software Requirements Specification (SRS) document and completed proposal formatting.
6	09/02/2026	9:00 AM	4:00 PM	• After completing proposal, spent the day validating industrial relevance of selected problem. • Watched simplified technical videos explaining how industrial gearboxes operate in manufacturing systems. • Focused on high-level understanding of: 1. Torque transmission 2. Heat generation 3. Mechanical wear progression • Observed example vibration graphs used in condition monitoring dashboards. • Understood basic difference between normal vibration patterns and abnormal spike patterns. • Studied how vibration monitoring is used in predictive maintenance without going into deep signal-processing mathematics. • Reflected on how vibration data would enter the "Industrial Sensor Data Layer" defined in architecture. • Evaluated whether deep vibration signal modeling is required or whether structured sensor datasets are sufficient. • Documented insights to ensure project remains AI-focused rather than mechanical-heavy.
7	10/02/2026	9:00 AM	4:00 PM	• Downloaded and explored NASA IMS bearing dataset to evaluate its suitability for the proposed predictive maintenance framework.

ay	Date	Time of Arrival	Time of Departure	List of activities performed by Student
				• Inspected folder structure and observed that dataset consists of multiple time-series vibration files. • Attempted to open and understand file format and data structure. • Realized that raw vibration signals require feature extraction before entering Machine Learning Layer. • Watched tutorials explaining how statistical features (RMS, mean, kurtosis) are extracted from vibration signals. • Evaluated preprocessing effort required to convert time-series signals into structured features. • Compared time required for signal preprocessing versus project timeline constraints. • Concluded that NASA dataset is academically strong but increases implementation complexity. • Documented reasoning for possibly shifting to structured dataset.
8	11/02/2026	9:00 AM	4:00 PM	• Explored structured industrial predictive maintenance datasets containing parameters such as: 1. Air temperature 2. Process temperature 3. Rotational speed 4. Torque 5. Tool wear • Inspected dataset format and confirmed it is tabular and directly usable for supervised learning. • Verified availability of failure labels for multi-class classification. • Compared structured dataset with NASA time-series dataset in terms of implementation feasibility. • Reflected on how structured sensor values will produce clearer SHAP explanations compared to abstract vibration signals. • Aligned dataset selection with SRS functional requirements • Structured initial implementation plan for: • Data loading

Day	Date	Time of Arrival	Time of Departure	List of activities performed by Student
				• Preprocessing • Model training • Explainability integration • Finalized decision to proceed with structured industrial dataset to maintain architectural clarity and timeline feasibility.
9	12/02/2026	9:00 AM	4:00 PM	• Spent time understanding what industrial gearbox sensor parameters should realistically look like. • Designed feature ranges for simulated data such as temperature, torque, rotational speed, and wear based on logical industrial assumptions. • Implemented Python code to generate a simulated dataset with multiple operating conditions. • Iteratively regenerated data multiple times to check realism of value ranges and label behavior. • Loaded the dataset into IBY.ipynb and performed multiple inspection checks (head, describe, class counts). • Modified simulation logic to avoid extreme or unrealistic values.
10	13/02/2026	9:00 AM	4:00 PM	• Explored relationships between simulated features using plots and summary statistics. • Checked data consistency, missing values, and distribution spread. • Tested multiple scaling methods before finalizing standard scaling. • Encoded multi-class failure labels and verified correctness. • Experimented with different train–test split ratios to observe class stability. • Refactored preprocessing code to ensure correct order of operations. • Re-ran notebook multiple times to ensure reproducibility.
11	16/02/2026	9:00 AM	4:00 PM	• Selected baseline ML models suitable for tabular industrial data. • Implemented and trained multiple models using the simulated dataset. • Ran experiments with different hyperparameter settings.

Day	Date	Time of Arrival	Time of Departure	List of activities performed by Student
				• Evaluated each model using accuracy, precision, recall, and F1-score. • Compared results across models to observe overfitting and generalization. • Re-trained selected model multiple times to confirm stability. • Analyzed confusion matrices to understand misclassification behavior.
12	17/02/2026	9:00 AM	4:00 PM	• Integrated SHAP to generate global and local explanations. • Analyzed feature influence across multiple predictions. • Implemented LIME for instance-level interpretability. • Generated LIME explanations for selected test samples. • Compared SHAP and LIME outputs to observe consistency and differences. • Adjusted feature presentation for clearer explanation visuals.
13	18/02/2026	9:00 AM	4:00 PM	• Re-ran complete pipeline including SHAP and LIME. • Verified end-to-end execution without errors. • Tested explanations on different samples to ensure consistency. • Cleaned redundant cells and improved notebook readability. • Prepared notebook outputs for screenshots and submission. • Noted limitations observed during explainability experiments.
14	19/02/2026	12:00 PM	4:00 PM	**Project Setup & Data Engineering** • Analyzed existing prototype and identified scope gaps — single model, random data, no deployment • Redesigned entire project into 5-module architecture covering data, ML, XAI, LLM and deployment • Wrote physics-informed data generation script using Archard's wear law, lubrication decay and thermal modeling • Generated 50,000 realistic gear sensor samples with proper feature correlations and AGMA-based fault labeling •

Day	Date	Time of Arrival	Time of Departure	List of activities performed by Student
15	20/02/2026	9:00 AM	4:00 PM	**Machine Learning Pipeline** • Applied SMOTE for class imbalance handling and configured 5-fold stratified cross validation • Trained and compared 5 ML models — Logistic Regression, Random Forest, Gradient Boosting, XGBoost and SVM • Gradient Boosting achieved 99.87% test accuracy and AUC 1.0000 and was auto-selected as best model • Trained separate Random Forest regression model for RUL prediction achieving R^2=0.9962 • Configured MLflow experiment tracking logging all parameters, metrics and artifacts across 9 runs
16	23/02/2026	9:00 AM	4:00 PM	**Explainability Layer** • Implemented SHAP TreeExplainer for global and local feature importance using XGBoost backend for multiclass support • Added LIME TabularExplainer to cross-validate SHAP findings with agreement percentage score • Trained Isolation Forest anomaly detection model on healthy gear data for early warning before fault classification • Generated fault progression timeline showing gear degradation from healthy to suspicious to minor to major fault •
17	24/02/2026	9:00 AM	4:00 PM	**LLM Copilot & Dashboard** • Built complete Streamlit dashboard with 6 tabs — AI Copilot, Model Comparison, SHAP+LIME, Anomaly Timeline, What-If Simulator and Report Generator • Integrated Groq API with LLaMA 3.3 70B for natural language chat interface, maintenance report generation and what-if scenario explanation • Implemented what-if simulator that reruns ML prediction with modified parameters and shows RUL delta instantly

Day	Date	Time of Arrival	Time of Departure	List of activities performed by Student
				• Added PDF report generator that auto-creates structured maintenance reports with root cause, action plan and cost analysis
18	25/02/2026	9:00 AM	4:00 PM	**UI Polish & Testing** • Completely redesigned UI with professional dark navy theme, gradient buttons, animated sensor bars and macOS-style chat window • Fixed markdown rendering in AI chat — bold, italic, bullet points and headings now render properly • Fixed SHAP and LIME sign interpretation for No Fault class so green bars correctly indicate healthy gear status • Verified complete system end-to-end including MLflow dashboard, AI Copilot, SHAP+LIME agreement, anomaly detection and report generator
19	26/02/2026	9:00 AM	4:00 PM	**Helical Prototype Completion** • XAI Engine Fix: Resolved the SHAP library compatibility error by patching the XGBoost booster attributes, enabling real-time feature impact visualization. • Anomaly Implementation: Integrated the Isolation Forest model to act as an early-warning system for detecting suspicious mechanical vibrations. • Final Testing: Verified the full pipeline from sensor slider input to AI Copilot response ensuring all reports and predictions are technically accurate.

ɪnature: ________________

(Student)

To be filled by Industry Mentor:

Student's overall evaluation: Excellent/Good/Satisfactory/Needs Improvement/Non-satisfactory

Specific Remarks : Excellent

Signature with Date:

To be filled by Internal Guide:

Student's overall evaluation: Excellent/Good/Satisfactory/Needs Improvement/Non-satisfactory

Specific Remarks :

Signature with Date: 27/2/26

G H Patel College of Engineering and Technology

The Charutar Vidya Mandal (CVM) University

Department of Information Technology, A.Y. 2025-26, Even Term

Industrial Internship (202080801)

Student's Activity Logbook

Enrollment No:	:	12202130501047
Name:	:	Om Nirajkumar Patel
Branch:	:	Computer Science and Design
Name & City of Industry	:	Tech Elecon PVT. LTD., Anand
Name and Designation of Industry Mentor	:	Mr. Satyam Raval, Deputy General Manager
Duration of Internship	:	From: 01/01/2026 To: 30/04/2026

Month:	March	Month Starting Date:	01/03/2026	Month Ending Date:	31/03/2026

Day	Date	Time of Arrival	Time of Departure	List of activities performed by Student
1	01/03/2026			**HOLIDAY**
2	02/03/2026	9:00 AM	4:00 PM	• Finalized the architectural shift from Streamlit to a **React + Vite** frontend and **FastAPI** backend. Designed a modular project structure separating data generation, ML training, XAI layers, and the web dashboard.
3	03/03/2026	9:00 AM	4:00 PM	• **Brief:** Implemented Bevel Gear physics-based data generator. • **Lead Evaluator:** Served as one of the 12 selected leads for main Elecon tasks; conducted weekly technical evaluations for interns' Data Science projects.
4	04/03/2026	9:00 AM	4:00 PM	• **Brief:** Refined ML pipeline using SMOTE and Gradient Boosting. • **Professional Evaluation:** Mentored 25 interns on AI/ML project workflows and verified their technical progress for the weekly department audit.
5	05/03/2026	9:00 AM	4:00 PM	• Integrated **SHAP TreeExplainer** and **LIME** for model transparency. Developed a cross-validation logic where if both XAI methods agree on a root cause, the system assigns a "high-trust" badge to the prediction.
6	06/03/2026	9:00 AM	4:00 PM	• Developed the **FastAPI** backend, creating 14 REST endpoints to handle real-time predictions, XAI artifacts, and history logging. Configured **CORS** settings to

Day	Date	Time of Arrival	Time of Departure	List of activities performed by Student
				allow secure communication between the React frontend and the Python server.
7	07/03/2026			**HOLIDAY**
8	08/03/2026			**HOLIDAY**
9	09/03/2026	9:00 AM	4:00 PM	• Bootstrapped the React application using **Vite** and established a global design system. Built the core **Main Dashboard** featuring interactive sensor sliders, real-time health gauges, and dynamic **Recharts** for RUL trends.
10	10/03/2026	9:00 AM	4:00 PM	• **Brief:** Configured RBAC and Super Admin (ompatel) access. • **Lead Evaluator:** Performed professional assessment of intern project architectures as part of the 12-man leadership team.
11	11/03/2026	9:00 AM	4:00 PM	• **Brief:** Engineered AI Copilot using LLaMA 3.3 and RAG. • **Professional Evaluation:** Assisted 25 interns in debugging AI/ML models and provided high-level technical guidance.
12	12/03/2026	9:00 AM	4:00 PM	• Created the **Reliability Data** and **Vibration PHM** pages. Integrated FMEA tables and **FFT spectral analysis** visualizations to help engineers monitor gear health through frequency domain indicators and Weibull distribution curves.
13	13/03/2026	9:00 AM	4:00 PM	• Developed a **What-If Optimizer** using the **Differential Evolution** algorithm. This allows operators to lock specific parameters, like load, while the AI finds the minimum adjustments needed to reach a safe health score.
14	14/03/2026			**HOLIDAY**
15	15/03/2026			**HOLIDAY**
16	16/03/2026	9:00 AM	4:00 PM	• Integrated **MLflow** for experiment tracking and model versioning. Logged all training runs, capturing hyperparameters

Day	Date	Time of Arrival	Time of Departure	List of activities performed by Student
				and AUC-ROC metrics, ensuring that every model deployment is backed by transparent and reproducible performance data.
17	17/03/2026	9:00 AM	4:00 PM	• **Brief:** Finalized Manufacturing QC and AGMA gear modules. • **Lead Evaluator:** Conducted the Tuesday lead review for student progress at Elecon; evaluated 25 diverse Data Science projects.
18	18/03/2026	9:00 AM	4:00 PM	• **Brief:** Performed E2E testing of the FastAPI inference engine. • **Professional Evaluation:**Validated intern project deliverables against industrial standards in a professional evaluator capacity.
19	19/03/2026	9:00 AM	4:00 PM	• Conducted **User Acceptance Testing (UAT)** with the industry mentor. Demonstrated the RBAC system, showing how the UI adapts based on the logged-in user's role, successfully hiding sensitive admin tools from technicians.
20	20/03/2026	9:00 AM	4:00 PM	• Optimized the frontend performance by implementing **Framer Motion** for smooth transitions and modularizing the **DashboardComponents** to ensure the 35KB CSS system loads efficiently across all industrial monitoring pages
21	21/03/2026			**HOLIDAY**
22	22/03/2026			**HOLIDAY**
23	23/03/2026	9:00 AM	4:00 PM	• Refined the **Staff Directory** and **Shift Schedule** pages for the admin role. Implemented search/filter functionality for employee management and a 3-shift scheduling visualization for plant supervisors.
24	24/03/2026	9:00 AM	4:00 PM	• **Brief:** Added Anomaly Detection using Isolation Forest • **Lead Evaluator:** Served as one of the 12 selected professional leads; conducted weekly technical evaluations for 25 interns' project milestones.

Day	Date	Time of Arrival	Time of Departure	List of activities performed by Student
25	25/03/2026	9:00 AM	4:00 PM	• **Brief:** Polished AI Report Generator and cost-benefit analysis sections. • **Professional Evaluation:** Provided expert mentorship to interns on Data Science workflows and industrial AI standards.
26	26/03/2026	9:00 AM	4:00 PM	• Completed the documentation for the **Elecon GearMind AI** system. Outlined the system architecture, file structure, and technical concepts like **AUC-ROC** and **SHAP** to assist the internal guide during evaluation.
27	27/03/2026	9:00 AM	4:00 PM	• Conducted a final stress test on the **FastAPI** server and **SQLite** database. Ensured that the 31-field history schema correctly logs every interaction, providing a full audit trail for industrial maintenance records.
28	28/03/2026			**HOLIDAY**
29	29/03/2026			**HOLIDAY**
30	30/03/2026	9:00 AM	4:00 PM	• Prepared the final presentation and **Demo Script**. Walked through the entire pipeline: from data generation and ML training to the live React dashboard with integrated AI Copilot and XAI transparency.
31	31/03/2026	9:00 AM	4:00 PM	• **Brief:** Final review of Activity Logbook and all project deliverables. • **Lead Evaluator:** Finalized professional evaluation reports for the cohort of 25 interns to conclude the March review cycle.

Signature: ____________

(Student)

To be filled by Industry Mentor:

Student's overall evaluation: Excellent/Good/Satisfactory/Needs Improvement/Non-satisfactory

Specific Remarks : He developed Elecon Germina AI & served as Evaluator professional monitoring 54 interns AI/ML projects.

Signature with Date: 6/5/2026

To be filled by Internal Guide:

Student's overall evaluation: Excellent/Good/Satisfactory/Needs Improvement/Non-satisfactory

Specific Remarks :

Signature with Date: 6/5/26

STUDENT'S ACTIVITY LOGBOOK

Enrollment No: : 12202130501047

Name: : Om Patel

Branch: : Computer Science and Design

Name & City of Industry : Tech Elecon Pvt. Ltd., Anand

Name and Designation of Industry Mentor : Mr. Satyam Raval, Deputy General Manager

Duration of Internship : **From:** 01/01/2026 **To:** 30/04/2026

Month:	4	Month Starting Date:	01/04/2026	Month Ending Date:	30/04/2026

Day	Date	Time of Arrival	Time of Departure	List of activities performed by Student
1	01/04/2026	9:00 AM	4:00 PM	• Set up the React.js + Vite project structure for the GearMind AI dashboard; configured folder layout, installed core dependencies including Recharts, Three.js, and Zustand. • Initialized the AuthContext provider and built the Login page with role-based credential validation for three shift operators and the super admin user.
2	02/04/2026	9:00 AM	4:00 PM	• Developed the Main Dashboard shell with a responsive tab navigation bar hosting seven primary module tabs; wired up global gear-type selector (Helical / Spur / Worm/Bevel) that persists across tabs. • Built the Gear Health tab — implemented animated circular health gauge, fault-status badge with colour coding, and confidence percentage display bound to the Flask /api/predict response.
3	03/04/2026	9:00 AM	4:00 PM	• Integrated sensor input sliders (Load, Torque, Vibration RMS, Temperature, Wear, Lubrication Index, Efficiency, Cycles) with range configuration fetched from /api/config per gear type. • Connected slider values to live prediction requests; added the RUL display card showing remaining cycles and estimated days alongside the health module output.

Day	Date	Time of Arrival	Time of Departure	List of activities performed by Student
4	06/04/2026	9:00 AM	4:00 PM	• Built the Three.js 3D gear visualisation component with involute tooth profile geometry; mapped gear health score to rotation speed and surface colour (steel-blue to amber to red). • Embedded the Three.js canvas into the Main Dashboard as an inline React component and resolved initial WebGL context initialisation issues on first render.
5	07/04/2026	9:00 AM	4:00 PM	• Developed the SHAP + LIME Explainability tab; integrated Plotly for the SHAP waterfall chart and a Recharts horizontal bar chart for LIME feature weights. • Wired the tab to the /api/shap and /api/lime endpoints; ensured chart updates re-trigger on each new prediction without redundant API calls.
6	08/04/2026	9:00 AM	4:00 PM	• Built the What-If Optimizer tab with parameter lock checkboxes and a Run Optimizer button; called the /api/what-if endpoint and rendered before/after metric cards. • Added a side-by-side parameter comparison bar chart and a radar chart overlaying original and optimised sensor profiles to visualise the recommended changes.
7	09/04/2026	9:00 AM	4:00 PM	• Developed the Vibration & PHM Analysis tab with Recharts multi-line charts showing session-level trends for vibration RMS, temperature, wear, and lubrication index. • Loaded fault progression data from the backend and rendered the historical fault progression chart with an anomaly detection overlay using Isolation Forest flags.
8	10/04/2026	9:00 AM	4:00 PM	• Built the Model Comparison tab displaying a benchmarking table of all trained classifiers (GBM, SVM, LR, RF, DT) with accuracy, F1, precision, and recall metrics per gear type. • Added a Recharts grouped bar chart to visualise model performance side by side; data sourced from the

Day	Date	Time of Arrival	Time of Departure	List of activities performed by Student
				model_comparison.json artifact via /api/config.
9	13/04/2026	12:00 PM	4:00 PM	• Started work on the Reliability & Fatigue Data page; designed the layout to present MTBF, design lifecycle hours, fatigue life reference values, and historical failure mode distributions. • Populated the section with gear-type-specific reliability statistics sourced from AGMA 2001 and Tech Elecon maintenance records; styled cards and data tables consistently with the dashboard theme.
10	14/04/2026	9:00 AM	4:00 PM	• Built the Design Parameters reference section within the Reliability page, listing gear-type-specific tooth geometry specifications, module range, pressure angle, and material grade data. • Added threshold reference tables for each sensor parameter per gear type, linking recommended operating bands to the health scoring logic used in the Main Dashboard.
11	15/04/2026	9:00 AM	4:00 PM	• Developed the Staff Directory page with a searchable table of maintenance and operations personnel; fields include name, role, department, contact, and assigned shift. • Built the Shift Schedule sub-view as a weekly calendar grid displaying operator-to-shift assignments for Shift A, B, and C; restricted both views to the super admin role.
12	16/04/2026	9:00 AM	2:00 PM	• Built the Manufacturing QC reference module displaying AGMA/ISO accuracy requirements, tooth profile tolerance bands, and surface roughness specifications per gear type. • Integrated the Cost Impact module with gear-type-specific repair, overhaul, and failure cost coefficients; rendered a comparison bar chart and savings summary card.
13	17/04/2026	9:00 AM	2:00 PM	• Completed the History Module which logs each prediction event in a session table recording timestamp, gear type, fault class, health score, confidence, and RUL.

Day	Date	Time of Arrival	Time of Departure	List of activities performed by Student
				• Added Recharts trend lines for health score and RUL across session history entries; confirmed smooth data flow from the Zustand global store into all history views.
14	20/04/2026	9:00 AM	4:00 PM	• Identified and fixed a WebGL context-loss bug in the Three.js component that caused a black screen when switching gear types rapidly; added an error boundary with automatic renderer re-initialisation. • Resolved a state synchronisation issue where switching gear type mid-session did not reset the slider defaults; updated the Zustand store to flush sensor values on gear-type change.
15	21/04/2026	9:00 AM	4:00 PM	• Fixed a CORS error on the /api/report endpoint that blocked PDF downloads in certain browser configurations; updated Flask-CORS settings to allow the content-disposition header. • Debugged an edge case where the What-If Optimizer returned invalid parameter sets when multiple sliders were locked simultaneously; added boundary enforcement in the backend objective function.
16	22/04/2026	9:00 AM	4:00 PM	• Corrected an off-by-one issue in the session History Module where the first prediction entry was occasionally duplicated in the log table after a gear-type switch. • Fixed LIME chart labels that displayed abbreviated column names instead of full sensor parameter names; updated the lime_tabular explainer to pass feature_names explicitly.
17	23/04/2026	9:00 AM	4:00 PM	• Addressed a layout overflow issue in the Staff Directory table on smaller viewport widths; applied responsive CSS adjustments to ensure horizontal scroll activates correctly. • Fixed the Shift Schedule calendar grid not updating when the admin edited shift assignments; corrected the React state mutation to trigger a proper re-render.

Day	Date	Time of Arrival	Time of Departure	List of activities performed by Student
18	24/04/2026	9:00 AM	4:00 PM	• Resolved a Recharts tooltip misalignment on the PHM Analysis trend chart that displayed incorrect sensor values on hover; corrected the data key mapping in the tooltip formatter. • Fixed a minor styling inconsistency in the Model Comparison bar chart where the Worm gear colour did not match the rest of the dashboard palette; updated the colour configuration object.
19	27/04/2026	9:00 AM	4:00 PM	• Began writing the project documentation covering system overview, technology stack, and module descriptions; drafted the Introduction and System Architecture sections. • Documented the ML pipeline — data preprocessing steps, model selection rationale, SHAP and LIME integration, and RUL estimation methodology — with supporting tables.
20	28/04/2026	9:00 AM	4:00 PM	• Documented all Flask API endpoints with request/response structure, parameter descriptions, and example payloads; wrote the System Design and Implementation chapters. • Wrote the Testing section with test case tables for the fault prediction module and API endpoints, including expected vs. actual output for all 18 test scenarios.
21	29/04/2026	9:00 AM	4:00 PM	• Completed the Conclusion and Discussion section covering project outcomes, problems encountered, limitations, and future enhancement recommendations. • Compiled the References list and appendices; performed a final review pass on the documentation for consistency, formatting, and accuracy against the actual system behavior.
22	30/04/2026	9:00 AM	4:00 PM	• Conducted a final walkthrough of all 13 dashboard modules with the industry mentor to verify the system against the original requirements; noted minor UI feedback for future iterations. • Completed internship activities; submitted the project documentation.

Signature: ____________
(Student)

To be filled by Industry Mentor:

Student's overall evaluation: Excellent/Good/Satisfactory/Needs Improvement/Non-satisfactory

Specific Remarks : Excellent Performance.

Signature with Date: 02/03/26

Tech Elecon Pvt. Ltd. Vallabh Vidyanagar

To be filled by Internal Guide:

Student's overall evaluation: Excellent/Good/Satisfactory/Needs Improvement/Non-satisfactory

Specific Remarks :

Signature with Date: 6/5/26

G H Patel College of Engineering and Technology

(A Charutar Vidya Mandal (CVM) University)

Department of Computer Engineering, A.Y. 2025-26, Even Term

Industrial Internship (202000801)

Student's Regularity / Attendance

Enrollment Number of Student	:	12202130501047
Name of Student	:	Om Patel
Division of Student	:	Computer Science & Design
Name & City of Industry	:	Tech Elecon Private Limited, Anand
Name and Designation of Industry Mentor	:	Mr Satyam Raval , Deputy General Manager
Total duration of Internship	:	**From:** 01/01/2026 **To:** 30/04/2026

Month & Year	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
January 2026			H	H				A		H	H			H	H	

Month & Year	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
January 2026	H	H						H	H	H					

Note:

1. Student should sign in the attendance column. Do not mark 'P'.
2. Holidays should be marked with 'H'. Absent should be marked as 'A' in attendance column.

Remarks by Industry Mentor (if any): ____________________

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Signature (with Date) and Seal: ____________________

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(Industry Mentor)

G H Patel College of Engineering and Technology

(A Charutar Vidya Mandal (CVM) University)

Department of Computer Engineering, A.Y. 2025-26, Even Term

Industrial Internship (202000801)

Student's Regularity / Attendance

Enrollment Number of Student	:	12202130501047
Name of Student	:	Om Patel
Division of Student	:	Computer Science & Design
Name & City of Industry	:	Tech Elecon Private Limited, Anand
Name and Designation of Industry Mentor	:	Mr Satyam Raval , Deputy General Manager
Total duration of Internship	:	From: 01/01/2026 To: 30/04/2026

Month & Year	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
February 2026	H						H	H						H	H	

Month & Year	17	18	19	20	21	22	23	24	25	26	27	28
February 2026					H	H					H	H

Note:

1. Student should sign in the attendance column. Do not mark 'P'.
2. Holidays should be marked with 'H'. Absent should be marked as 'A' in attendance column.

Remarks by Industry Mentor (if any): ____________________

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Signature (with Date) and Seal: ____________________

(Industry Mentor)

G H PATEL COLLEGE OF ENGINEERING AND TECHNOLOGY
(A CHARUTAR VIDYA MANDAL (CVM) UNIVERSITY)
DEPARTMENT OF COMPUTER ENGINEERING, A.Y. 2025-26, EVEN TERM
INDUSTRIAL INTERNSHIP (202000801)

GCET G H PATEL COLLEGE OF ENGINEERING & TECHNOLOGY

CVM UNIVERSITY

STUDENT'S REGULARITY / ATTENDANCE

Enrollment Number of Student	: 12202130501047
Name of Student	: Om Patel
Division of Student	: Computer Science & Design
Name & City of Industry	: Tech Elecon Private Limited, Anand
Name and Designation of Industry Mentor	: Mr Satyam Raval , Deputy General Manager
Total duration of Internship	: From: 01/01/2026 To: 30/04/2026

Month & Year	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
March 2026	H						H	H						H	H	

Month & Year	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
March 2026					H	H						H	H		

Note:
1. Student should sign in the attendance column. Do not mark 'P'.
2. Holidays should be marked with 'H'. Absent should be marked as 'A' in attendance column.

Remarks by Industry Mentor (if any): Perfect Punctuality & highly Professional

Tech Elecon Pvt. Ltd.

Signature (with Date) and Seal
(Industry Mentor)

G H PATEL COLLEGE OF ENGINEERING AND TECHNOLOGY
(A CHARUTAR VIDYA MANDAL (CVM) UNIVERSITY)
DEPARTMENT OF COMPUTER ENGINEERING, A.Y. 2025-26, EVEN TERM
INDUSTRIAL INTERNSHIP (202000801)

STUDENT'S REGULARITY / ATTENDANCE

Enrollment Number of Student : 12202130501047
Name of Student : Om Patel
Division of Student : Computer Science & Design
Name & City of Industry : Tech Elecon Private Limited, Anand
Name and Designation of Industry Mentor : Mr Satyam Raval, Deputy General Manager
Total duration of Internship : **From:** 01/01/2026 **To:** 30/04/2026

Month & Year	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
April 2026				H	H						H	H				

Month & Year	17	18	19	20	21	22	23	24	25	26	27	28	29	30
April 2026		H	H						H	H				

Note:
1. Student should sign in the attendance column. Do not mark 'P'.
2. Holidays should be marked with 'H'. Absent should be marked as 'A' in attendance column.

Remarks by Industry Mentor (if any): ____________________

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Signature (with Date) and Seal: 03/05/26 Tech Elecon Pvt. Ltd.
(Industry Mentor)

FEEDBACK FORM BY INDUSTRY MENTOR
202000801 - INDUSTRIAL INTERNSHIP

CVM UNIVERSITY
Aegis: Charutar Vidya Mandal (Estd.1945)

INDUSTRY MENTOR'S DETAILS: -

Mentor's Name: Mr. Satyam Raval
Mentor's Company/Organization: Tech Elecon Private Limited
Contact Details: (M): 7066207670
(Email ID): ssraval1@techelecon.com

STUDENT'S DETAILS: -

Enrolment No.: 12202130501047
Student's Name: Om Patel
Dates of Internship: 01/01/2026 To 30/04/2026
Branch: Computer Science and Design
College Name: GCET

EVALUATION MATRIX OF INTERN/STUDENT: -

(Assessment of the Intern/Student by showing the recurrence with which you noticed the accompanying ways of behaving)

Evaluation Aspect/s	Excellent	Good	Satisfactory	Needs Improvement	Non-satisfactory
Student's Regularity and Punctuality (towards assigned tasks and attendance during the period of the internship)	✓				
Work Quality (refers to value of work delivered by the student, accuracy, and competency level)	✓				
Technical Knowledge (refers to knowledge, clarity of fundamentals, and the latest development)	✓				
Problem Solving Skills (refers to the involvement to find best alternative for any problem)	✓				
Professional Attitude (refers to the quality of work and recognizes accountability)	✓				
Communication Skills (refers to the way of expression, communication, and presentation of idea / thought)	✓				
Interpersonal Relationship (refers to the ability to work harmoniously with superiors, subordinates – as a team)	✓				

Overall Performance: - Excellent/Good/Satisfactory/Needs Improvement/Non-satisfactory

Specific Remarks, if any:

Tech Elecon Pvt. Ltd. Vallabh Vidyanagar

02/05/2026

Name and Signature of the Industry Mentor/Supervisor with Company Seal